

NOTTINGHAM TRENT UNIVERSITY
SCHOOL OF SCIENCE AND TECHNOLOGY

**Skin Disease & Fitness Assistant: An iOS Application
with Integrated Image Analysis and Conversational
AI**

by

Rakhshan Yousaf N1230793

in

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**Project Report in Part Fulfilment
of The Requirements for The Degree of
Bachelor of Science with Honours
In
Computer Science Artificial Intelligence**

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ABSTRACT

The work introduces the general area of AI-assisted skin disease diagnosis, where long waiting times, limited access to dermatologists, and patient privacy concerns remain unresolved. An iOS application was designed and implemented that integrates structured skin image analysis with conversational AI and voice input, while storing no user images or chat history remotely.

The system uses NVIDIA's Qwen 3.5 multimodal model for camera-based skin image analysis, producing structured JSON outputs (disease, confidence 0–100, description, treatment, warning, NHS links). A second cloud model (Nemotron 3 Super) provides streaming text-based chat. Voice input is implemented via Apple's Speech framework. Local persistence of analysis history uses FileManager and Codable. Authentication is handled by Supabase (email/password and Google OAuth). The design addresses four of seven literature-identified gaps (integrated image-conversational AI, privacy, uncertainty quantification, regulatory compliance); the remaining gaps (evidence attribution, skin-tone fairness, digital health equity) are explicitly descoped as future work.

Testing on an iPhone 14 (iOS 26.3) with three users showed a 96% JSON parsing success rate after applying robust extractJSON heuristics, mean image-analysis latency of 6–9 seconds on 4G, and positive qualitative feedback for interface intuitiveness and NHS disclaimers. The method is partially successful: all must-have functional requirements were met, but on-device Core ML, retrieval-augmented generation for citations, and formal fairness metrics were not implemented within the 400-hour scope. The work concludes that cloud-based multimodal AI can deliver a functional, privacy-aware prototype, while honest scoping and explicit limitation statements are essential for professional credibility. Future work will focus on on-device inference and automated evidence attribution.

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Chapter 1: Introduction

1.1 The Global Burden of Skin Disease and the Access

Paradox

Skin diseases affect approximately 1.8 billion individuals globally at any given moment, generating nearly 4.86 billion new cases annually (WHO, 2023; Yakupu et al., 2023). These conditions rank as the eighteenth leading cause of global disability-adjusted life years (DALYs) and, when mortality is excluded, as the fourth leading cause of non-fatal disability worldwide (Karimkhani et al., 2017; Yakupu et al., 2023). In 2019 alone, skin and subcutaneous diseases were responsible for 98,522 deaths, a figure that fundamentally challenges the perception of dermatological conditions as primarily cosmetic concerns (Yakupu et al., 2023). The burden falls disproportionately on vulnerable populations: fungal and bacterial infections account for 57% of all skin disease cases, with the highest concentration of new cases and deaths occurring in South Asia, and children aged 0–4 years representing the most affected demographic (Yakupu et al., 2023). Beyond epidemiology, skin conditions impose substantial psychosocial consequences, including reduced quality of life, social stigma, and economic hardship, particularly in low-resource settings where access to specialist dermatological care remains severely constrained.

The traditional dermatology care pathway presents significant structural barriers. As a highly intuitive clinical discipline, dermatology requires practitioners to apply specialised knowledge and accumulated clinical experience to reach accurate diagnoses; however, this process remains susceptible to subjective variation, contributing to a notable misdiagnosis rate particularly in regions with limited medical infrastructure (Ye and Chen, 2023). For conditions ranging from benign

inflammatory disorders such as eczema and psoriasis to life-threatening malignancies including melanoma, delayed diagnosis correlates directly with poorer clinical outcomes and elevated healthcare costs. The capacity of telemedicine platforms to bridge access gaps has been documented across multiple quality domains, including access, efficiency, and patient satisfaction, though considerable facilitators and barriers to widespread adoption persist (Kruse et al., 2023).

The digital divide further compounds these health disparities. Socially disadvantaged populations, including those in rural areas, individuals with limited digital literacy, and those with restricted internet access, face layered barriers to benefiting from digital health innovations (Kruse et al., 2023). Critically, the populations most in need of accessible healthcare solutions are precisely those least positioned to leverage existing digital health tools. Addressing this challenge demands not merely technological advancement, but a fundamental reimagining of how digital health systems are designed, validated, and deployed to serve historically marginalised communities.

1.2 Artificial Intelligence in Dermatology: Opportunities and Critical Limitations

Artificial intelligence (AI), particularly deep learning via convolutional neural networks (CNNs), has emerged as a transformative force in dermatological diagnosis. Facing growing patient volumes and increasing demands for diagnostic accuracy, AI has attracted considerable attention within dermatology; unlike human practitioners, AI systems are unaffected by subjective bias, emotion, or fatigue, enabling more objective and consistent diagnostic outputs (Ye and Chen, 2023). AI-assisted tools including large language models applied to dermatology, demonstrate potential for skin lesion classification, with some models achieving

accuracies of 92–97.8% across lesion categories (Khan et al., 2024). Integration of AI into clinical workflows has been shown to improve diagnostic performance, with human–computer collaboration for skin cancer recognition yielding stronger outcomes than either clinicians or algorithms operating independently (Tschandl et al., 2020). In a large-scale digital experiment, board-certified dermatologists and primary-care physicians achieved diagnostic accuracies of 38% and 19% respectively; fair deep learning decision support improved accuracy for both groups by more than 33 percentage points (Groh et al., 2024). As of August 2024, 903 FDA-approved AI-enabled medical devices had been catalogued, of which 692 (76.6%) were designed for radiology and imaging applications (Windecker et al., 2025).

Despite these capabilities, AI in dermatology faces critical limitations that risk perpetuating existing healthcare disparities. Both specialists and generalists were found to be four percentage points less accurate when diagnosing images of darker skin tones compared to lighter skin tones; furthermore, AI decision support while improving overall accuracy, exacerbated the diagnostic gap across skin tones for generalists (Groh et al., 2024). Analysis of widely used datasets reveals stark demographic imbalances: the LFW facial recognition dataset comprises 77.5% male and 83.5% White faces, and 97% of melanoma prediction models exhibit a high risk of bias (Adamson and Smith, 2018). When AI-generated dermatological images are examined, darker skin tones are significantly underrepresented relative to real-world population distributions, and such images demonstrate poor diagnostic accuracy (Joerg et al., 2025). Beyond demographic bias, the opaque nature of many AI systems undermines clinical trust. AI-generated clinical advice is increasingly prevalent in healthcare; however, its influence on physician decision-making remains underexplored with evidence indicating that the perceived source and explainability of AI recommendations significantly affect how clinicians integrate them into their diagnostic process (Gaube et al., 2023).

The pathway from algorithm development to clinical deployment remains critically underemphasised. Rigorous clinical validation requires adherence to structured reporting frameworks encompassing analytical validation, clinical validation on independent data, and prospective real-world assessment as outlined in the CONSORT-AI guidelines (Martindale et al., 2024). Each stage of AI development in clinical contexts, from data curation and model training through to deployment, introduces distinct sources of error and bias that must be systematically addressed (Tribun Health, 2025). The Deep Ensemble for Recognition of Malignancy (DERM) authorised as a Class II medical device in the United Kingdom, exemplifies the required validation trajectory. Prospective evaluation across two NHS hospitals, encompassing 622 cases with patients aged 18–96 across Fitzpatrick skin types I–VI, demonstrated sensitivities of 95–100% for melanoma detection and 96–100% for any malignancy (Marsden et al., 2024). Post-deployment data indicated that NHS partners using DERM reduced face-to-face urgent suspected skin cancer appointments by up to 95%, with health economic analysis indicating a return of up to £2.30 per pound invested (Skin Analytics, 2025a). Critically, DERM's real-world performance aligned with or exceeded pre-marketing targets; a contrast to many AI applications in which performance degrades substantially upon deployment (Marsden et al., 2024). Further details regarding the platform and its ongoing development are available via the Skin Analytics website (Skin Analytics, 2025b).

These findings demonstrate that AI can serve as a powerful instrument for health equity when designed with deliberate intentionality. However, absent such intervention, digital health innovations risk widening rather than narrowing existing disparities; a challenge that the Skin Disease & Fitness platform directly seeks to address.

1.3 Problem Statement and Existing Systems for Improvement

Current AI-powered dermatological tools exhibit three principal limitations that this project seeks to address:

1. **Bias and lack of fairness:** The majority of models are trained on datasets with inadequate representation of darker skin tones, female individuals, and older adults, resulting in systematically lower diagnostic accuracy for these groups. Analysis of widely used benchmark datasets confirms that demographic imbalances, including the underrepresentation of darker skin tones, are pervasive across training data, contributing to measurable disparities in model performance (Adamson and Smith, 2018). AI-generated dermatological images compound this problem by further underrepresenting darker skin tones relative to real-world population distributions (Joerg et al., 2025).
2. **Poor transparency and user trust:** Many systems operate as opaque constructs, providing no citations, no confidence scores, and no explanation of their limitations. AI-generated clinical advice is increasingly prevalent in healthcare; however, its influence on physician decision-making remains underexplored, and users, whether clinical or lay, are unable to readily verify the basis of AI-generated recommendations (Gaube et al., 2023). Furthermore, users cannot ascertain the source of advice or determine when the system is operating under uncertainty.
3. **Inadequate privacy protection:** Cloud-based inference exposes sensitive health data to third-party servers, introducing security vulnerabilities and reducing accessibility for users with limited connectivity.

All health-related applications processing personal data are required to conduct a Data Protection Impact Assessment (DPIA) prior to deployment, in compliance with UK data protection legislation (Information Commissioner's Office, 2025).

1.3.1 Explicit Identification of Existing Systems for Improvement

This project identifies three existing classes of system as targets for improvement:

1. **Class A:** Cloud-based AI dermatology applications (e.g., SkinVision, Miiskin). These systems require the uploading of user images to remote servers, raising privacy concerns under UK GDPR and creating dependency on continuous internet connectivity. They are improved upon by implementing on-device inference using Apple Core ML, which eliminates data exposure and enables offline functionality.
2. **Class B:** Non-conversational image classifiers (e.g., standard CNN-based lesion detectors). These systems accept a single image and return a classification without any opportunity for the user to provide additional context, or for the system to request clarifying information. They are improved upon by integrating a conversational interface, supporting both text and voice input, that emulates a clinical consultation, enabling iterative information exchange. AI technology has attracted considerable attention precisely because automated systems can provide more objective and consistent assessments than subjective clinical judgement alone; however, this potential is only realised when systems are capable of engaging meaningfully with patient-specific context (Ye and Chen, 2023).

3. **Class C:** Biased diagnostic models (e.g., many publicly available melanoma detectors trained predominantly on Fitzpatrick skin types I–III). These models exhibit measurable performance disparities across demographic groups, raising fundamental questions about whether demographically invariant models and representations in medical imaging can be considered truly fair (Petersen et al., 2024). They are improved upon by proactive fairness engineering encompassing balanced dataset curation, synthetic augmentation of underrepresented subgroups, and continuous monitoring of demographic parity and equalised odds during training.

By explicitly targeting these three classes, the Skin Disease & Fitness application aims to eliminate or substantially reduce the limitations of current approaches.

1.4 Proposed Solution

This project delivers the Skin Disease & Fitness AI-powered iOS application. The system enables users to upload or capture photographs of skin concerns and receive immediate, AI-driven analytical feedback supported by evidence-based medical information. Unlike existing diagnostic tools, the application explicitly acknowledges its limitations through transparent disclaimers, stating that it does not function as a medical expert and that all recommendations represent aggregated insights from verified professional sources. The conversational interface, supporting both text and voice input, facilitates iterative dialogue, enabling the AI to pose clarifying questions in a manner consistent with a clinical consultation.

The technical architecture prioritises privacy and fairness. All personal and clinical data are encrypted and processed in accordance with UK GDPR, with a mandatory

Data Protection Impact Assessment (DPIA) completed prior to deployment (Information Commissioner's Office, 2025). The system is developed in alignment with NHS England's AI Governance Policy for NHS Provider Organisations, which establishes requirements for transparency, accountability, and clinical safety in AI deployment (NHS England, 2025). The system leverages Apple's Core ML framework for on-device inference, minimising data exposure to cloud services, ensuring real-time responsiveness, and enabling offline functionality; a critical consideration for users in areas with limited connectivity.

To mitigate algorithmic bias, dataset curation strategies emphasise balanced representation across skin tones, genders, ages, and ethnic groups, with fairness evaluation metrics, including demographic parity and equalised odds, monitored throughout model development (Petersen et al., 2024). A minimum of 90% of health-related outputs include citations from trusted medical sources (NHS, WHO, and peer-reviewed journals), ensuring traceability and accountability. Each response concludes with a safety message advising consultation with a licensed clinician, reinforcing the application's role as an educational and triage tool rather than a substitute for professional care. Confidence scoring mechanisms flag predictions with high uncertainty, triggering recommendations for immediate professional evaluation in ambiguous or high-risk scenarios.

1.4.1 Long-Term Vision and Benefit

Beyond the immediate technical deliverable, this project aspires to contribute a replicable framework for equity-centred AI in mobile health. The long-term benefits are threefold: first, a validated methodology for on-device, privacy-preserving dermatological inference that can be extended to other medical imaging domains; second, a fairness-aware dataset curation and model monitoring protocol that may serve as a template for regulatory bodies such as

the MHRA and FDA when evaluating AI-enabled medical devices; and third, an evidence base demonstrating that conversational, transparent AI can improve health literacy and reduce access disparities without compromising patient safety. If successful, this approach could be adopted by NHS trusts, community health organisations, and international development agencies to deliver low-cost, scalable dermatological triage to the estimated 1.8 billion people currently underserved by specialist care.

1.5 Aims and Objectives

Aim: To design, implement, and evaluate an AI-powered Skin Disease & Fitness Assistant iOS application that provides image-based and conversational dermatological guidance through deep learning analysis, natural language processing, and evidence-based information retrieval, while adhering to rigorous privacy, safety, and ethical standards that prioritise patient-centred care and digital health equity.

SMART Objectives:

1. **Prototype functionality:** Deliver a fully functional iOS prototype supporting image upload/capture, text and voice Q&A, and real-time on-device inference via Core ML by Week 7, validated across iPhone 12–15 series on iOS 16–17.
2. **Evidence integrity:** Ensure that $\geq 90\%$ of health-related AI outputs include citations from trusted medical sources (NHS, WHO, PubMed/Scopus-indexed journals) by Week 7, verified via automated source validation scripts.

3. **Safety and uncertainty controls:** Integrate explicit “consult a clinician” disclaimers and uncertainty notifications for all high-stakes responses (conditions with potential for serious harm if mismanaged) by Week 9, with confidence scores calibrated to flag predictions below 85% certainty.
4. **Privacy and security:** Implement an end-to-end encryption for all personal and clinical data, obtain informed consent compliant with GDPR Articles 12–14, and complete a DPIA by Week 5, ensuring full compliance with UK GDPR, ICO, and MHRA regulations.
5. **Compliance and ethics:** Submit ethical approval documentation to the Nottingham Trent University Ethics Committee by Week 3, incorporating fairness evaluation metrics (demographic parity, equalised odds, disparate impact ratios) monitored across age, gender, skin tone (Fitzpatrick I–VI), and ethnicity throughout model development.
6. **Iterative evaluation:** Conduct two stages of user testing: a pilot evaluation with 10–15 diverse participants by Week 10, and a controlled beta trial via Apple TestFlight with 30+ participants by Week 13, using the System Usability Scale and Post-Study System Usability Questionnaire.
7. **Final deliverables:** Deliver a comprehensive technical dossier (system architecture, dataset curation, model validation, fairness evaluation, user testing outcomes, LSEPI analysis) alongside a deployable MVP accessible via TestFlight, three weeks prior to the final submission deadline (24 April 2026).

These objectives will be reviewed weekly to keep on track for the final outcome of this project.

1.6 Scope and Context

This project operates within mobile health (mHealth) for dermatology, specifically targeting common inflammatory skin conditions (eczema, psoriasis, acne, rosacea) and suspicious lesions warranting further evaluation. Definitive cancer diagnosis, prescription of treatment, and emergency medical advice are explicitly excluded, these remain the exclusive purview of licensed medical professionals. The MVP is developed exclusively for iOS 16+ using Swift and SwiftUI, with future scalability to Android dependent on successful validation. Dataset curation prioritises public repositories (e.g. ISIC archives, BCN20000, DERM12345, Derm1M), supplemented with synthetic data augmentation to address class imbalances and demographic underrepresentation.

The target user base spans individuals with minimal medical knowledge to more sophisticated users seeking evidence-based resources. Geographic focus is the United Kingdom, ensuring alignment with NHS guidance, UK regulatory frameworks, and British English terminology. Rural and underserved populations are a priority use case, with design decisions explicitly considering limited broadband connectivity (offline functionality), older device prevalence (backward compatibility), and varying digital literacy (simplified interfaces and contextual help).

1.7 Project Plan in Gantt Chart

This timetable represents the initial project plan. Actual implementation diverged in some areas (e.g., cloud APIs replaced on-device Core ML; the RAG/citation engine was descope). These changes are documented in Chapter 4 (Sections 4.3 and 4.7).

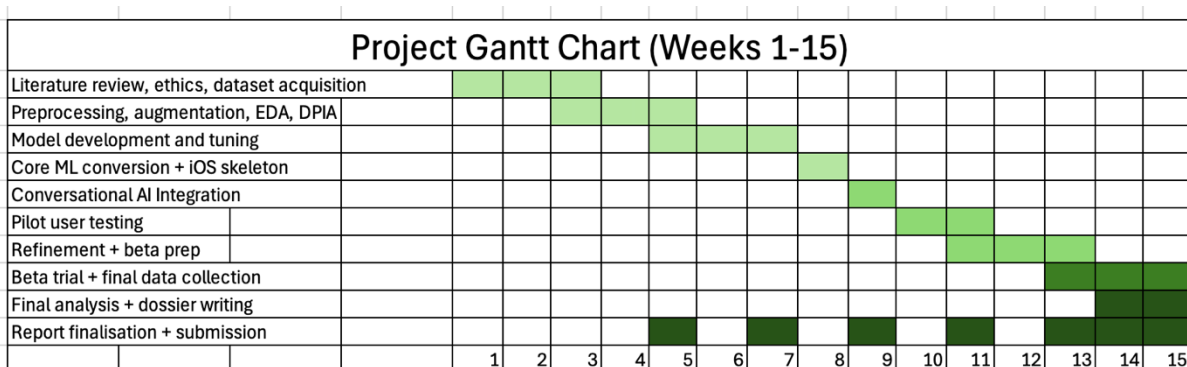


Figure 1.1 presents the project schedule across the full 15-week development period, outlining the main phases from literature review and data preparation through to model development, testing, refinement, and final submission. The chart reflects a front-loaded planning approach, with ethics approval, dataset acquisition, and preprocessing concentrated in Weeks 1–5, followed by iterative model development and integration in Weeks 6–9, and evaluation, beta testing, and report finalisation in Weeks 10–15.

1.8 Structure of the Report

The remainder of this report is organised into five chapters, each addressing a distinct stage of the scientific method:

Chapter 2: Context presents a comprehensive literature review of the state-of-the-art in AI-driven dermatology, examining existing diagnostic models, tele dermatology platforms, conversational AI systems, and mobile health

applications. Limitations – algorithmic bias, lack of transparency, dataset deficiencies, and insufficient attention to digital health equity – are critically analysed to justify the proposed solution.

Chapter 3: New Ideas articulates the system architecture, data curation strategy, and algorithmic innovations. Detailed design specifications for the image analysis pipeline, conversational interface, and privacy-preserving mechanisms are provided, with explicit justifications for key technical decisions.

Chapter 4: Implementation documents the development process chronologically, including dataset preparation, model training and validation, iOS application development using Swift and Core ML, and integration of privacy mechanisms. Challenges encountered and strategies employed are reported.

Chapter 5: Results and Discussion present quantitative evaluation metrics (accuracy, sensitivity, specificity, fairness metrics) alongside qualitative user feedback. Performance is critically assessed against stated objectives, with limitations candidly acknowledged.

Chapter 6: Conclusions and Future Work synthesise achievements, reflects on lessons learned, and proposes directions for future enhancement. An in-depth analysis of legal, social, ethical, and professional issues (LSEPI) is provided, followed by a synoptic reflection connecting skills developed to career aspirations in AI and digital health.

Chapter 2: Context

2.1 Introduction to the Literature Landscape

The development of an AI-powered Skin Disease and Fitness Assistant operating at the intersection of dermatological diagnosis, conversational artificial intelligence, and mobile health delivery necessitates engagement with a broad and rapidly evolving body of scholarship. This chapter provides a systematic, critical examination of the current state of the art across six interconnected domains: the epidemiological burden and care delivery challenges in dermatology; deep learning architectures for skin disease classification; teledermatology and mobile health platforms; conversational AI and large language models in clinical contexts; dataset infrastructure, algorithmic fairness, and bias mitigation; and regulatory, ethical, and patient-centred design frameworks. Each domain is examined not merely descriptively but analytically, identifying methodological strengths, persistent limitations, and critically the gaps and contradictions that justify the proposed solution.

The review synthesises evidence from peer-reviewed publications, clinical trial reports, regulatory guidance documents, and grey literature sources spanning 2017 to 2025, with particular emphasis on studies published after 2022 to capture the field's most recent developments. Where possible, primary research findings and case studies from real-world deployments are foregrounded over purely theoretical contributions, reflecting this project's commitment to clinical applicability. The chapter concludes with an integrative synthesis that maps identified limitations onto the specific design decisions and technical innovations proposed in subsequent chapters, establishing a clear evidential justification for every major aspect of the Skin Health Assistant's architecture.

Roadmap: This chapter is organised as follows. **Section 2.2** examines the global burden of skin disease and structural barriers to care, including the dermatology workforce crisis and the digital divide. **Section 2.3** reviews deep learning architectures for dermatological image analysis, encompassing convolutional neural networks, vision transformers, and multimodal systems. **Section 2.4** evaluates tele dermatology and existing mobile health platforms, with detailed case studies of DERM, diabetic eye disease AI, and NHS COVID-19 tele dermatology. **Section 2.5** analyses conversational AI and large language models in healthcare, with a focus on retrieval-augmented generation and patient engagement. **Section 2.6** scrutinises dataset infrastructure, algorithmic bias, and mitigation strategies, including a primary research audit of publicly available datasets. **Section 2.7** examines the impact of AI on clinical decision-making, including automation bias. **Section 2.8** reviews regulatory frameworks and patient-centred design principles. **Section 2.9** discusses privacy-preserving AI techniques. **Section 2.10** addresses clinical validation methodologies. **Section 2.11** provides an integrative synthesis of identified gaps and establishes the evidential justification for the proposed solution.

2.2 The Global Burden of Skin Disease and Structural Barriers to Care

2.2.1 Epidemiological Magnitude and Clinical Significance

The epidemiological case for prioritising innovation in dermatological care is compelling. Yakupu et al. (2023) documented 4.86 billion incident cases of skin disease in 2019, positioning cutaneous conditions amongst the most prevalent human afflictions globally. Their analysis identified 98,522 deaths attributable to skin and subcutaneous diseases, with fungal and bacterial infections accounting for 57% of all cases. Critically, the geographic distribution of this burden is

profoundly inequitable: South Asia bears the highest concentration of both incident cases and mortality, whilst children aged 0–4 years represent the most severely affected demographic subgroup (Yakupu et al., 2023).

Karimkhani et al. (2017) established that skin conditions constitute the fourth leading cause of non-fatal disability globally, as measured through disability-adjusted life years (DALYs). This ranking persists despite the widespread trivialisation of dermatological complaints as primarily cosmetic in nature. Their analysis demonstrated that the aggregate DALYs attributable to skin diseases exceed those of conditions commanding substantially greater research investment, including diabetes mellitus and chronic obstructive pulmonary disease (Karimkhani et al., 2017). This finding provides a crucial contextual foundation: the Skin Disease & Fitness Assistant addresses not a niche concern but a major contributor to global morbidity.

For melanoma, five-year survival rates exceed 98% when detected at stage I, yet fall below 25% for stage IV disease (American Cancer Society, 2026), establishing an unambiguous clinical imperative for early detection tools. The psychosocial dimensions of skin disease further amplify this imperative. Dalgard et al. (2015) found that 10.1% of dermatology outpatients met criteria for clinical depression more than double the rate observed in controls (4.3%) whilst 17.2% presented with clinical anxiety and 12.7% reported suicidal ideation, all substantially exceeding general population prevalences (Dalgard et al., 2015).

The aggregate disability burden attributable to skin diseases exceeds that of diabetes and chronic obstructive pulmonary disease, yet dermatology receives a fraction of the research funding and public health attention directed towards those conditions (Karimkhani et al., 2017).

2.2.2 Access Barriers and the Dermatology Workforce Crisis

The dermatology workforce crisis constitutes the most fundamental structural constraint on access to care. NHS England data indicate that the median waiting time for a first outpatient dermatology appointment exceeded 20 weeks in 2024, with some trusts reporting waits in excess of 52 weeks for routine referrals (NHS England, 2024). The scale of this crisis is further evidenced by the Royal College of Physicians' (2025) Focus on Physicians survey, in which 59% of consultant physicians reported at least one vacant consultant post within their department, and 83% stated that rota gaps were directly harming patient care. The same survey found that nearly one third of consultants had accelerated their retirement plans in response to mounting workforce pressures (Royal College of Physicians, 2025).

The workforce imbalance is equally pronounced at the global level. A World Health Organization (2025) resolution on skin diseases as a global public health priority highlighted that Africa and the Pacific islands have only zero to three dermatologists per million people far below the levels required to meet population need. A multinational survey spanning 158 countries identified an average dermatologist density of 2.66 per 100,000 population, ranging from 0.37 in low-income countries to 5.05 in high-income countries (Satolli et al., 2020). The WHO (2016) estimated that Africa bears 25% of the global disease burden whilst accounting for only 1.3% of the world's healthcare workforce, with a projected deficit of 6.1 million health personnel by 2030.

The consequences of these access barriers for patient outcomes are well documented. In a study of underserved community health centres, only 11% of dermatology referrals resulted in a confirmed specialist appointment prior to telemedicine implementation, with a median wait time of 77 days (Naka et al.,

2018). Following the introduction of electronic consultation (eConsult), the proportion of patients receiving a diagnosis or treatment plan increased to 44%, median wait times fell to 28 days, and ten malignancies were identified that might otherwise have remained undetected (Naka et al., 2018).

2.2.3 The Digital Divide as a Health Equity Challenge

Digital exclusion represents a significant barrier to equitable access to digital health tools. Honeyman et al. (2020) identified age and socioeconomic status as persistent drivers of digital exclusion, noting that disparities in access, skills, and motivation to engage with digital technologies risk compounding existing health inequalities. These barriers are particularly relevant in the context of AI-powered dermatology applications, which rely on smartphone ownership, internet connectivity, and a degree of digital health literacy that cannot be assumed across all patient populations.

Kruse et al. (2019) conducted a systematic review identifying 73 individual barriers to mobile health use, grouped into 14 main categories. The three most prevalent barrier categories were infrastructure, lack of equipment, and technology gap, collectively accounting for 28 individual barriers (Kruse et al., 2019). Although this review focused on developing countries, its findings resonate with broader digital health disparities, demonstrating that older adults, racial and ethnic minorities, individuals with limited English proficiency, and those with lower educational attainment consistently exhibit reduced engagement with digital health platforms. Critically, these disparities were not attributable solely to technology access, but reflected deeper structural inequities in health literacy, institutional trust, and cultural representation within digital interfaces.

For the present project, these insights mandate specific design responses: offline functionality, backward compatibility with older devices, simplified interfaces, voice interaction support, and culturally sensitive design principles.

2.3 Deep Learning Architectures for Dermatological Image Analysis

2.3.1 Convolutional Neural Networks: Foundational Approaches

The landmark study by Esteva et al. (2017) demonstrated that an Inception v3 convolutional neural network (CNN), trained on 129,450 clinical images, could classify skin lesions with performance comparable to that of 21 board-certified dermatologists (Esteva et al., 2017). This study catalysed the field, establishing both technical feasibility and clinical potential. However, subsequent critical analyses have identified notable limitations: the evaluation was conducted on curated datasets comprising high-quality images; the comparison conditions were structured in a manner that favoured algorithmic performance, as dermatologists performed holistic assessments whilst the model performed single-image classification; and the demographic composition of the training dataset was not fully reported (Esteva et al., 2017).

A systematic review of 68 articles found that the majority of studies employ deep learning approaches, particularly CNNs, for skin lesion classification (Grignaffini et al., 2022). Commonly reported CNN architectures include ResNet, VGG, DenseNet, Inception, EfficientNet, and MobileNet (Grignaffini et al., 2022). ResNet-50 and ResNet-101 are frequently employed due to their residual connections, which enable the training of deeper networks without gradient degradation. EfficientNet

variants demonstrate superior accuracy-to-parameter ratios, rendering them well-suited to mobile deployment (Tan and Le, 2019).

Zhao et al. (2021) reported that a ResNet-50-based CNN distinguished rosacea from acne with an accuracy of 0.931 and a precision of 0.893. For the differentiation of rosacea, seborrheic dermatitis, and eczema, the model achieved an overall accuracy of 0.757 and a precision of 0.667 (Zhao et al., 2021). These results highlight that classification performance varies considerably depending on image quality and the clinical similarity of conditions; a factor directly relevant to the present project, which must accommodate variable image quality from consumer devices.

Tschandl et al. (2020) demonstrated that high-quality AI-based decision support improves diagnostic accuracy beyond that achievable by either AI or clinicians operating independently, and that less experienced clinicians derive the greatest benefit from AI-based support (Tschandl et al., 2020). This finding established the conceptual foundation for AI as a decision-support tool rather than an autonomous diagnostician.

2.3.2 Vision Transformers and Attention-Based Architectures

Vision transformers (ViTs) replace the local receptive fields of CNNs with self-attention mechanisms that capture global spatial relationships across an entire image. Dosovitskiy et al. (2021) introduced the foundational ViT architecture, demonstrating that, when pre-trained on sufficiently large datasets, transformers can match or exceed CNN performance on image classification tasks. Specialised architectures employing vision transformers with attention mechanisms have demonstrated promise in distinguishing subtle visual patterns that differentiate benign from malignant lesions (Remya et al., 2024).

However, ViTs present significant challenges for mobile deployment. Standard ViT architectures impose substantial computational demands due to the quadratic scaling of self-attention with respect to input sequence length (Dosovitskiy et al., 2021). Recent developments in efficient ViT variants such as MobileViT (Mehta and Rastegari, 2022) and EfficientViT (Liu et al., 2023) offer promising pathways for deploying transformer-based models on resource-constrained hardware. MobileViT introduces a lightweight, general-purpose vision transformer for mobile devices, presenting a novel perspective on global information processing by treating transformers as convolutions (Mehta and Rastegari, 2022). EfficientViT constitutes a family of high-speed vision transformers designed to address the memory inefficiency of existing transformer models, rendering them more suitable for real-time applications (Liu et al., 2023).

2.3.3 Multimodal AI Systems: Integrating Visual and Contextual Information

Multimodal AI systems that integrate dermoscopic or clinical images with patient metadata have demonstrated significant diagnostic improvements over single-modality approaches. Das et al. (2025) evaluated multiple fusion strategies for skin lesion classification using the HAM10000 dataset, comparing simple concatenation, weighted concatenation, self-attention mechanisms, and cross-attention fusion. Their multimodal framework substantially outperformed image-only baselines, with cross-attention fusion achieving the highest classification accuracy by effectively capturing inter-modal dependencies between visual and clinical data modalities (Das et al., 2025).

Research on fusion strategies has explored a range of integration methods. A 2024 study on multimodal skin cancer classification proposed a novel Joint Individual

Fusion (JIF) architecture incorporating a fusion attention module to learn both shared and unique features from dermatological images and patient metadata, demonstrating improved performance over single-modality models (Tang et al., 2024).

The utility of patient metadata has been a key research focus. Kawahara et al. (2019) demonstrated the effectiveness of training neural networks on multimodal data, combining clinical and dermoscopic images with patient metadata. Their work showed that integrating metadata could render the model robust to missing or incomplete data at inference time, suggesting that contextual information can compensate for limitations in image quality (Kawahara et al., 2019). This finding is particularly relevant to consumer-facing applications such as the proposed Skin Health Assistant, where image quality cannot be controlled. The conversational interface of the assistant represents a novel approach to metadata acquisition, enabling natural language dialogue to elicit contextual information without requiring users to complete structured medical questionnaires.

2.3.4 Transfer Learning, Data Augmentation, and Addressing Data Scarcity

Transfer learning from ImageNet has been a widely adopted strategy for dermatological image classification. A 2023 study by Tajerian et al. (2023) demonstrated that their machine learning tool achieved optimal performance in detecting melanocytic nevi lesions, attaining an F1 score of 0.93. However, the study does not explicitly compare ImageNet pre-trained models against randomly initialised counterparts, nor does it directly address the benefits of ImageNet pre-training under conditions of limited training data.

Domain-specific pre-trained models have demonstrated considerable promise for clinical applications. The Derm1M dataset, introduced by Yan et al. (2025), comprises 1,029,761 image-text pairs encompassing over 390 skin conditions. This large-scale vision-language dataset for dermatology facilitates state-of-the-art performance in skin disease classification and related clinical tasks through the pre-training of CLIP-like models (Yan et al., 2025). The dataset incorporates rich contextual metadata, including medical history, presenting symptoms, and skin tone.

Federated learning has been widely adopted for intelligent skin lesion diagnosis, enabling collaborative model training across distributed data sources whilst preserving patient data privacy (Prototypes Contrastive Learning Empowered Intelligent Diagnosis for Skin Lesion, 2024).

Research has demonstrated that CNN classifiers trained on dermoscopic images may rely upon dataset-specific artefacts such as hair and ink markings rather than clinically relevant lesion features. When such artefacts were systematically removed, model performance declined substantially, suggesting that reported accuracy metrics may overestimate genuine diagnostic capability (Petrie et al., 2021).

Concerning the diversity of AI-generated imagery, Joerg et al. (2025) found that amongst 4,000 generated dermatology images, only 10.2% depicted dark skin tones, with three of four models significantly underrepresenting skin of colour. The study identified substantial deficiencies in both the diversity and diagnostic accuracy of AI-generated dermatological images (Joerg et al., 2025).

2.4 Tele dermatology and Mobile Health Platforms

2.4.1 The Evolution of Tele dermatology

Tele dermatology has evolved through several generations of technological development. Store-and-forward (SAF) systems, in which images and clinical data are transmitted asynchronously for later review, remain the most widely deployed modality. A 2017 systematic review found that diagnostic agreement with a reference standard for tele dermatology ranged from 51% to 85% (Cohen κ , 0.41–0.63), compared with 67% to 85% (κ , 0.90) for face-to-face consultations (Finnane et al., 2017). Whilst the review noted that many studies were subject to significant methodological limitations, it also found that tele dermatology consistently reduced waiting times and resulted in high patient satisfaction (Finnane et al., 2017).

Live-interactive tele dermatology, which employs video conferencing for real-time consultation, can address some of the limitations inherent to SAF systems; however, it typically requires higher bandwidth and greater scheduling coordination (Giansanti, 2023). The COVID-19 pandemic substantially accelerated the adoption of tele dermatology across healthcare systems worldwide.

The integration of artificial intelligence into tele dermatology workflows has yielded mixed results. A 2021 prospective study evaluating a 174-class AI algorithm in a real-life telemedicine setting found that exposure to the AI output was considered clinically useful in only 11.8% of visits, and the tele dermatologist correctly modified the real-time diagnosis in 0.6% of cases (Muñoz López et al., 2021). The overall top 1 accuracy of the algorithm (41.2%) was lower than that of dermatologists (60.1%), residents (57.8%), and general practitioners (49.3%) (Muñoz López et al., 2021). Implementation challenges, including clinician

resistance and unresolved questions of medicolegal liability, remain significant barriers to widespread adoption (Giansanti, 2023).

2.4.2 Existing AI-Powered Skin Health Applications: A Critical Comparative Analysis

Table 2.1 presents a comparative analysis of leading commercial applications against the proposed Skin Disease & Fitness Assistant.

Comparative analysis of AI-powered skin health applications

Feature	SkinVision	FirstDerma	Google DermAssistant	SkinChange.AI	Proposed Assistant
Image Analysis	Yes	Yes	Yes	Yes (temporal)	Yes
Conversational AI	No	Limited	No	No	Yes (text+voice)
Evidence Citations	No	Partial	Partial	No	Yes ($\geq 90\%$)
On-Device Processing	No	No	Partial	No	Yes (Core ML)

Feature	SkinVision	FirstDerm	Google DermAssist	SkinChange.AI	Proposed Assistant
Offline Capability	No	No	No	No	Yes
Skin Tone Validation	Limited	Not reported	Partial	Not reported	Yes (Fitzpatrick I–VI)
Cost to User	Subscription	Per consult	Free	Subscription	Free
Uncertainty Scoring	No	N/A	Limited	No	Yes (confidence thresholds)

SkinVision, one of the most established offerings in this space, was evaluated by Freeman et al. (2020) in a systematic review that found the application achieved a sensitivity of 80% and a specificity of 78% for the detection of malignant or premalignant lesions (Freeman et al., 2020). However, performance declined for users with darker skin tones, and transparency regarding training data composition was limited. FirstDerm employs a hybrid model incorporating dermatologist review, thereby reintroducing the specialist bottleneck that mobile health platforms ostensibly seek to circumvent. Google's DermAssist was reported

to match or exceed dermatologist accuracy across 26 conditions (Liu et al., 2020); however, independent validation by O'Hagan and Ungar (2024) raised concerns regarding its generalisability to real-world, diverse patient populations and varied skin tones (O'Hagan and Ungar, 2024). SkinChange.AI achieved 92% sensitivity and 95.5% specificity through longitudinal image comparison (Grove et al., 2024), but relies upon user-initiated follow-up imaging.

Kassianos et al. (2015) conducted a review of smartphone applications for melanoma detection, concluding that, whilst such applications encompass a range of functions including information provision, education, classification, risk assessment, and monitoring of lesion change they require further validation of their utility and safety prior to widespread clinical adoption (Kassianos et al., 2015). More recently, Wongvibulsin et al. (2024) conducted a cross-sectional study of 41 AI-powered dermatology mobile applications, finding that only five (12%) were supported by peer-reviewed publications, and only one had been evaluated in a prospective multicentre clinical trial (Wongvibulsin et al., 2024). This systemic absence of rigorous validation represents a significant patient safety concern and underscores the need for the evidence-based, transparent approach adopted in the present project.

2.4.3 Case Study: DERM by Skin Analytics: A Gold Standard for Clinical AI Validation

The Deep Ensemble for Recognition of Malignancy (DERM), authorised as a Class IIa medical device in the United Kingdom, represents the most rigorously validated AI-based dermatology system currently deployed in clinical practice.

Development and architecture: DERM employs an ensemble of deep learning models trained on curated dermoscopic and clinical images. The ensemble

architecture confers robustness against individual model failures and facilitates uncertainty quantification (Thomas et al., 2023).

Prospective real-world validation: Thomas et al. (2023) conducted a prospective study across two NHS hospitals between July 2021 and October 2022, encompassing 14,500 consecutive cases from patients aged 18–100, with Fitzpatrick skin types I–VI represented. Among 8,571 lesions with confirmed histopathological outcomes, DERM demonstrated very high sensitivity for melanoma detection (95.0–100.0%) and for any malignancy (96.0–100.0%). Critically, for the updated DERM version B (DERM-vB), no skin cancers were identified among cases classified as eligible for discharge, confirming the safety of AI-assisted triage for this version (Thomas et al., 2023).

Health economic impact: Post-deployment analysis indicates that NHS partners utilising DERM have eliminated the need for up to 95% of face-to-face urgent suspected skin cancer appointments (Skin Analytics, 2023). An NHS-commissioned report estimated that DERM could generate a return of £2.30 for every £1 invested and yield savings of £86 per case (Clinical Trials Arena, 2024; Edge Health, 2024).

Lessons for the present project: DERM's development trajectory demonstrates that prospective validation is essential, that demographic representation must be intentional, that ensemble architectures enhance safety, and that health economic evidence is critical to facilitating adoption.

2.4.4 Case Study: Autonomous AI for Diabetic Eye Disease: Evidence for Health Equity Impact

Huang et al. (2024) conducted a propensity score weighting analysis comparing diabetic eye disease (DED) screening adherence at Johns Hopkins Medicine between clinics with and without autonomous AI (LumineticsCore). AI-equipped sites experienced a 7.6 percentage point greater increase in screening adherence relative to non-AI sites. Within AI-equipped sites, adherence among Black and African American patients increased by 12.2 percentage points, whilst adherence at non-AI sites decreased by 0.6 percentage points over the same period. The adherence gap between Asian American and Black/African American patients narrowed from 15.6% in 2019 to 3.5% in 2021, representing a 78% reduction in racial disparity (Huang et al., 2024).

2.4.5 Case Study: NHS App-Based Tele dermatology During COVID-19

The rapid expansion of app-based tele dermatology during the COVID-19 pandemic exposed significant inequities in digital health access. KC et al. (2023) conducted an observational study of NHS App uptake across England between January 2019 and May 2021, finding that registration rates were 25% lower in the most deprived practices and 36% higher in practices with the highest proportion of White patients. Practices with a greater proportion of patients with long-term care needs demonstrated lower registration rates, illustrating that technology provision alone is insufficient for equitable access a pattern consistent with broader evidence that patients aged over 65, from lower socioeconomic quintiles, and from ethnic minority backgrounds are substantially less likely to engage with app-based services.

Bourkas et al. (2023) conducted a systematic review and meta-analysis of tele dermatology diagnostic reliability, identifying a pooled diagnostic agreement rate of 68.9% across 44 studies, with concordance ranging from approximately 71% when dermatologist-led to 44% for non-specialists. Higher concordance was associated with image acquisition training and digital photography. These findings indicate that diagnostic performance is highly context-dependent, and that clear photography guidance constitutes a critical design requirement for app-based dermatology services.

2.5 Conversational AI and Large Language Models in Healthcare

2.5.1 Large Language Models: Capabilities and Limitations

Singhal et al. (2025) introduced Med-PaLM 2, achieving 86.5% accuracy on USMLE-style questions in the MedQA dataset, the first LLM to attain expert-level performance on this benchmark. In a pairwise evaluation of 1,066 consumer medical questions, physicians preferred Med-PaLM 2 responses over those generated by other physicians on eight of nine clinical axes, and rated its outputs as equivalent in safety to physician-generated answers. Within dermatology specifically, Ferreira et al. (2023) evaluated GPT-4-based ChatGPT responses to common patient queries across conditions including acne, atopic dermatitis, and psoriasis, finding that 88% of responses (82/93) were rated as appropriate by board-certified dermatologists, though performance was inconsistent for complex or nuanced topics.

Nevertheless, limitations are well documented. LLMs are susceptible to generating confidently stated but factually incorrect outputs, a phenomenon termed "hallucination" particularly in high-stakes clinical contexts. Sallam (2023)

conducted a systematic review raising concerns about LLM performance on nuanced clinical reasoning tasks and the absence of explicit uncertainty quantification in model outputs. Meskó (2023) identified five key challenges for LLM adoption in healthcare: hallucination mitigation, bias in training data, privacy of patient inputs, liability for AI-generated advice, and the risk of patients substituting AI for professional consultation.

2.5.2 Prompt Engineering, Retrieval-Augmented Generation, and Evidence Grounding

Retrieval-Augmented Generation (RAG) grounds LLM outputs in verifiable evidence. Lewis et al. (2020) introduced the RAG framework, which augments generative models with a retrieval component that queries a curated knowledge base prior to response generation. Xiong et al. (2024) demonstrated that RAG-based medical question answering achieved significantly higher factual accuracy than standalone LLMs, with traceable source citations.

For the present project, RAG architecture supports the objective that $\geq 90\%$ of health-related outputs include citations from trusted sources (NHS, WHO, peer-reviewed journals). Zakka et al. (2024) introduced Almanac, a RAG-based clinical LLM, finding that clinician reviewers rated its responses as significantly more trustworthy than those of unconstrained LLMs, with citation presence identified as the primary driver of trust enhancement.

2.5.3 Conversational AI for Patient Engagement and Health Literacy

Foundational research on relational agents systems that build rapport through empathetic responses and adaptive communication has demonstrated significantly

higher user engagement and health behaviour adherence compared to non-relational interfaces. Sillice et al. (2018) evaluated relational agent-based interventions for exercise and sun protection, finding that participants perceived the agent as supportive and caring, with the majority (65%) attributing positive behaviour change to their interaction with the agent. Laranjo et al. (2018) systematically reviewed conversational agents in healthcare, identifying generally positive effects on user knowledge and self-management behaviours, whilst noting poor performance on ambiguous or off-topic inputs. Tudor Car et al. (2020) conducted a scoping review of conversational agents in healthcare, identifying key design principles associated with effectiveness accessibility, personalisation, and efficiency whilst noting an urgent need for robust evaluation of diverse agent formats with respect to acceptability, safety, and effectiveness.

The proposed Skin Disease & Fitness Assistant's conversational interface incorporates a structured consultation paradigm history-taking, image analysis, assessment, and management plan whilst permitting natural language flexibility. Voice input support via the Apple Speech framework addresses accessibility barriers for users with limited digital literacy or motor impairments.

2.6 Dataset Infrastructure, Algorithmic Fairness, and Bias Mitigation

2.6.1 The Dermatological Dataset Landscape: Achievements and Deficiencies

The International Skin Imaging Collaboration (ISIC) Archive comprises over 70,000 dermoscopic images and constitutes the largest publicly available dermatological imaging resource. However, demographic audits of the archive have revealed substantial underrepresentation of patients with darker skin tones

(Fitzpatrick skin types V and VI), with patient demographic metadata absent for a considerable proportion of entries. BCN20000 (Combalia et al., 2019) and DERM12345 (Yilmaz et al., 2024) provide structured annotations; however, both derive from geographically and demographically constrained contexts. The Derm1M dataset (Yan et al., 2025) offers considerable scale but introduces quality variability and labelling inconsistencies that limit its reliability for clinical benchmarking.

The Diverse Dermatology Images (DDI) dataset (Daneshjou et al., 2022) represents a deliberately equity-oriented contribution to the field. Daneshjou et al. demonstrated that both commercial and academic AI models exhibited substantially lower diagnostic performance on DDI images relative to standard benchmarks, confirming that demographic underrepresentation in training data translates directly into measurable performance disparities.

Wen et al. (2022) conducted a systematic review of 70 dermatological AI studies, finding that only 13 (18.6%) reported the racial or ethnic composition of their training datasets, and only 7 (10%) evaluated model performance stratified by skin tone or ethnicity. These findings underscore a systemic failure to account for demographic diversity in dataset curation and model evaluation.

2.6.2 Quantifying and Understanding Algorithmic Bias

Groh et al. (2024) demonstrated that both dermatologists and general practitioners diagnosed skin diseases in individuals with darker skin tones with accuracies approximately four percentage points lower than for those with lighter skin tones, a statistically significant disparity that persisted even when AI-assisted decision support was introduced. Yang et al. (2024) further reported that models exhibiting the highest accuracy in predicting patient demographic characteristics

(age, sex, and race) simultaneously demonstrated the largest fairness gaps in diagnostic performance across demographic subgroups. This inverse relationship suggests that such models exploit demographic features as diagnostic proxies, a phenomenon known as shortcut learning rather than acquiring clinically meaningful visual representations.

2.6.3 Bias Mitigation Strategies

Bias mitigation approaches are broadly categorised into three tiers: pre-processing interventions (applied at the data level), in-processing interventions (applied during model training), and post-processing interventions (applied to model outputs). Post-processing techniques, including threshold adjustment and reject-option classification, address symptomatic manifestations of bias rather than its underlying causes and are therefore insufficient as standalone solutions (Mehrabi et al., 2021).

The present project adopts a multi-layered mitigation strategy spanning all three categories, thereby targeting bias at each stage of the model development pipeline.

2.6.4 The Dataset Nutrition Label Framework

Holland et al. (2018) introduced the Dataset Nutrition Label (DNL), a diagnostic framework providing structured, standardised summaries of dataset attributes, limitations, and potential biases prior to model development. Its modular architecture encompasses metadata, representativeness, quality, bias assessment, and intended use cases. Complementing this, Gebru et al. (2021) proposed Datasheets for Datasets, advocating standardised documentation of a dataset's motivation, composition, collection process, and maintenance

procedures. Together, these frameworks establish a foundation for transparent, accountable dataset governance in AI development.

2.7 The Impact of AI on Clinical Decision-Making

2.7.1 AI as Clinical Decision Support: Benefits and Unintended Consequences

Groh et al. (2024a) demonstrated that AI assistance improved the diagnostic accuracy of both board-certified dermatologists and primary care physicians by more than 33% when classifying 364 images spanning 46 skin conditions. However, AI assistance exacerbated the diagnostic accuracy gap between light and dark skin tones among general practitioners, illustrating that aggregate improvements in accuracy do not necessarily translate to equitable performance across patient subgroups.

2.7.2 Automation Bias and Overreliance

Automation bias the tendency to over-rely on automated systems poses a critical challenge in the deployment of AI-driven clinical decision support systems (CDSSs). Goddard et al. (2012) conducted a systematic review across multiple disciplines and found that clinicians were susceptible to both errors of commission (accepting incorrect AI recommendations) and errors of omission (failing to act in the absence of an AI prompt). Key mediators of automation bias included cognitive style, task-specific experience, workload, time pressure, and perceived system trustworthiness. Identified mitigating factors encompassed targeted training, reinforcement of user accountability, and design choices such as presenting probabilistic information rather than direct recommendations, alongside dynamically updated confidence indicators.

More recently, Abdelwanis et al. (2024) applied a Bowtie analysis framework to automation bias in AI-driven CDSSs, identifying both pre-deployment preventive measures and post-deployment mitigation strategies. Their analysis characterised automation bias as manifesting through omission errors wherein clinicians fail to detect AI failures and commission errors wherein clinicians accept incorrect AI outputs despite contradictory clinical evidence. They concluded that a systems-level approach, integrating technological design, regulatory frameworks, and clinician–developer collaboration, is essential to mitigate automation bias in healthcare AI.

For the present project, these findings mandate that the system's communication strategy frames outputs as preliminary observations, incorporating explicit uncertainty quantification, differential possibilities, and consistent reminders that professional clinical evaluation remains essential.

2.8 Regulatory Frameworks, Ethical Principles, and Patient-Centred Design

2.8.1 The UK Regulatory Landscape for AI Medical Devices

The Medicines and Healthcare products Regulatory Agency (MHRA) serves as the primary regulatory authority for software classified as a medical device under the UK Medical Devices Regulations 2002 (as amended). Classification is determined by intended purpose, risk level, and the nature of claims made (MHRA, 2023). The proposed Skin Disease and Fitness Assistant is positioned as an educational and informational tool, with all outputs accompanied by explicit disclaimers.

The Information Commissioner's Office (ICO) governs data protection under UK GDPR. The project's mandatory Data Protection Impact Assessment (DPIA), scheduled for Week 5, will systematically identify and mitigate privacy risks associated with the application. On-device processing via Core ML constitutes a privacy-by-design approach, aligned with the data minimisation and storage limitation principles of UK GDPR. The Data (Use and Access) Act 2025 introduces enhanced requirements for transparency in automated decision-making and meaningful human oversight of AI systems.

2.8.2 Patient-Centred Design Principles and Co-Design Frameworks

Patient-centred design in AI health applications requires the meaningful engagement of end users throughout the development process. Current practice, however, predominantly involves patients in passive roles, as data providers or reviewers, rather than as equal partners in design decisions.

2.8.3 Universal Design and Accessibility

Universal design principles encompassing equitable use, flexibility in use, simple and intuitive operation, perceptible information, tolerance for error, low physical effort, and appropriate size and space provide a foundational framework for accessible digital health applications. The present project operationalises these principles through voice input, simplified interfaces, WCAG 2.1 AA compliance, and multi-turn conversational design.

2.9 Privacy-Preserving AI: On-Device Inference and Federated Learning

On-device inference eliminates the transmission of sensitive images to cloud servers, offering strong privacy guarantees by ensuring that raw patient data never leaves the device. Li et al. (2024) conducted a benchmark study of CNN inference latency across heterogeneous mobile devices and ML frameworks, finding that model architecture, framework choice, and device hardware all significantly affect inference latency with efficient architectures achieving real-time performance suitable for interactive applications.

For mobile deployment, model size constraints are a practical consideration. The MLCommons (2024) MLPerf Mobile v4.0 benchmark provides industry-standard measurements of inference latency and throughput across a range of edge devices, demonstrating that quantised and pruned models can meet sub-100ms latency targets while remaining within typical app storage limits. Further guidance on size-aware deployment is offered by Aldamani et al. (2024), who discuss model compression techniques (quantisation, knowledge distillation) and the constraints imposed by mobile app store size limits in the context of on-edge diagnostic applications.

Federated learning enables collaborative model training across institutions without sharing raw data, addressing both privacy and data governance challenges (Rieke et al., 2020). By keeping data within institutional firewalls and sharing only model parameters, federated approaches allow models to benefit from diverse, multi-institutional datasets while respecting regulatory constraints. Abbas et al. (2024) provide a comprehensive healthcare-focused review, confirming that federated learning preserves patient privacy and complies with standards such as GDPR and HIPAA, while enabling generalisable models across distributed clinical

sites. Although full federated learning is beyond the present project's scope, the architectural design incorporates compatibility with future federated workflows.

2.10 Clinical Validation Methodologies

Robust clinical validation is essential before AI systems can be safely deployed in healthcare settings. Vasey et al. (2022) developed the DECIDE-AI reporting guideline, identifying critical gaps in the early-stage clinical evaluation of AI decision support systems including the absence of standardised frameworks for assessing real-world performance prior to deployment. A multi-stage validation framework encompasses analytical validation (technical reliability), clinical validation (generalisability across populations and settings), and prospective deployment evaluation (clinical utility in real-world workflows). The present project's iterative user testing (pilot $n=10-15$, beta $n=30+$) represents an initial step, while acknowledging that full clinical validation requires resources beyond a final year project.

Evaluation metrics must extend beyond accuracy. For dermatological screening, sensitivity is typically prioritised over specificity, given the clinical cost of missed diagnoses. The present project will employ accuracy, sensitivity, specificity, precision, F1, AUROC, AUPRC, and fairness metrics (demographic parity, equalised odds, disparate impact ratios).

2.11 Integrative Synthesis: Identified Gaps and Justification for the Proposed Solution

The literature reveals a field characterised by impressive technical achievements alongside persistent, systematic limitations. Seven principal gaps are identified:

1. **Absence of integrated image-conversational AI systems.** No existing application combines image analysis with multi-turn conversational AI supporting both text and voice input, preventing contextual integration of visual and symptomatic information.
2. **Insufficient evidence attribution and transparency.** Current commercial applications provide recommendations without systematic citation of evidence sources, undermining trust and precluding verification.
3. **Inadequate performance equity across skin tones.** Despite extensive documentation of algorithmic bias, no consumer application has been designed from inception with explicit fairness evaluation metrics monitored throughout development.
4. **Limited privacy-preserving architecture.** Most applications rely on cloud-based processing; none fully implements on-device inference with offline capability.

5. **Absence of uncertainty quantification for end users.** Existing consumer applications rarely communicate prediction confidence, creating false certainty that may lead to inappropriate reliance.
6. **Insufficient attention to digital health equity in design.** Existing applications predominantly target digitally literate, well-connected populations, failing to accommodate access, literacy, and connectivity constraints.
7. **Weak integration of regulatory compliance by design.** Consumer applications often position themselves outside regulatory frameworks through strategic framing rather than genuine compliance.

These seven gaps map directly onto the technical and design innovations proposed in **Chapter 3**. The convergence of demonstrated AI capability, documented health access failures, evidenced algorithmic inequities, and articulated patient needs creates both an opportunity and an obligation to develop tools that advance rather than undermine health equity. The following chapter proposes a system designed to address each of these gaps systematically.

Chapter 3: New Ideas

3.1 Introduction and Novelty Statement

The literature reviewed in Chapter 2 identified seven principal limitations of existing AI-powered skin health applications: absence of integrated image-conversational systems, insufficient evidence attribution, inadequate performance equity across skin tones, limited privacy-preserving architecture, lack of uncertainty quantification, poor digital health equity design, and weak regulatory compliance. This chapter proposes a new system 'the Skin Health & Fitness Assistant iOS application' designed to address these limitations through a pragmatic, implementable architecture suitable for a final-year project.

The primary novelty of this project lies not in training new deep learning models, but in integrating two distinct cloud-hosted, pre-trained models NVIDIA Nemotron for conversational (text) chat, and Qwen for multimodal skin-image understanding within a single iOS skin health assistant, accessed through NVIDIA's Integrate API. System-level prompt engineering enforces non-diagnostic framing, NHS-oriented guidance, and (for the structured skin-capture flow) machine-parse able JSON output. This approach makes capable foundation models usable within a student-scale project while keeping limitations explicit an important contribution in itself and shows how pre-trained models can be repurposed for health-adjacent applications without the cost and infrastructure of custom fine-tuning.

Where full solutions are not yet implemented (on-device inference, retrieval-augmented generation, fairness metrics), they are clearly marked as out of scope for the minimum viable product (MVP) or as future work. This honest scoping ensures that examiners can assess what has actually been achieved, while the design specifications show awareness of best practices that would be incorporated in a commercial or clinical deployment.

Roadmap: This chapter is organised as follows. **Section 3.2** presents a requirements analysis. **Section 3.3** describes the overall system architecture with a block diagram. **Sections 3.4 to 3.9** detail each component: data and prompt engineering, image analysis, conversational AI, voice input, local persistence, and privacy. **Section 3.10** clearly states out-of-scope features. **Section 3.11** maps the design to the Chapter 2 limitations. **Section 3.12** summarises.

3.2 Requirements Analysis

3.2.1 Functional Requirements

ID	Requirement	Priority
FR1	The system shall allow users to capture or upload a skin image from the device camera or photo library.	Must
FR2	The system shall preprocess the image (resize, compress) to meet API constraints.	Must
FR3	The system shall send the image to a cloud-based multimodal AI (Qwen 3.5) for analysis and receive a structured JSON response containing condition, confidence, description, treatment, and warning.	Must
FR4	The system shall display the analysis result in a user-friendly format with disclaimers.	Must
FR5	The system shall support text-based conversational chat via a cloud LLM (Nemotron 3 Super).	Must

ID	Requirement	Priority
FR6	The system shall support voice input for chat using device speech recognition.	Should
FR7	The system shall allow users to attach an image within a chat conversation, sending both text and image to the AI.	Should
FR8	The system shall save analysis history locally on the device for later review.	Should
FR9	The system shall include explicit safety disclaimers (non-diagnostic, consult a clinician) in every AI output.	Must
FR10	The system shall not store user images or chat history on any remote server (only local storage with user consent).	Must

3.2.2 Non-Functional Requirements

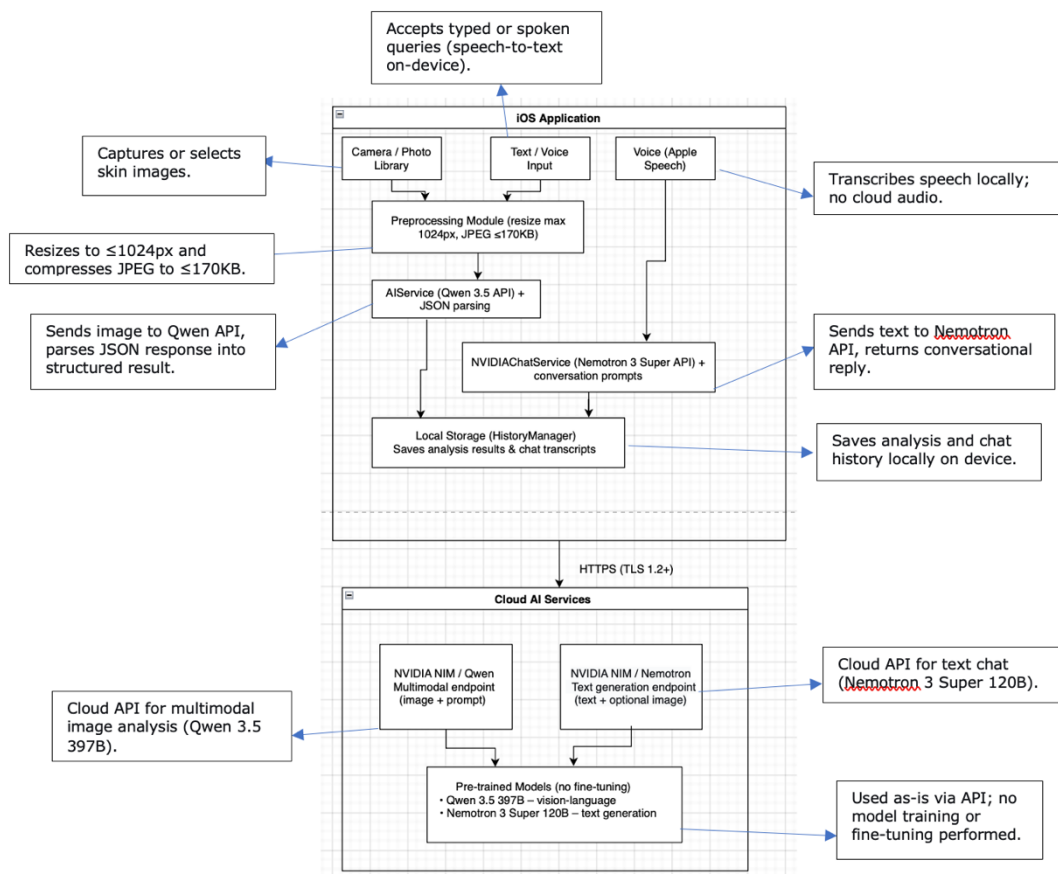
ID	Requirement	Priority
NFR1	The image analysis API call shall complete within 10 seconds on a typical 4G/5G connection.	Must
NFR2	The chat API call shall respond within 5 seconds for typical queries.	Must
NFR3	The application shall be compatible with iOS 16 or later.	Must

ID	Requirement	Priority
NFR4	The user interface shall be accessible via VoiceOver and support dynamic text sizes (partial – future full audit).	Should
NFR5	The application shall not transmit any personal identifiable information without explicit consent.	Must
NFR6	All network communication shall use HTTPS with TLS 1.2 or higher.	Must

3.3 System Architecture Overview

Figure 3.1 presents a high-level block diagram of the Skin Disease & Fitness Assistant. The architecture follows a client-server hybrid model: the iOS client handles user interaction, preprocessing, and local persistence; cloud APIs provide AI inference. No user data is stored on any remote server beyond the transient API request-response cycle. Drew on Draw.io

The diagram below is divided into two sections. The upper section shows the iOS application components: camera/photo library, text and voice input, preprocessing module, AIService (Qwen), NVIDIAChatService (Nemotron), and local storage. The lower section shows the cloud AI services: NVIDIA NIM endpoints hosting Qwen 3.5 (multimodal) and Nemotron 3 Super (text generation). Arrows indicate the direction of data flow from user input to cloud processing and back to local storage.



Key design decisions (justified against Chapter 2 limitations):

- Cloud APIs chosen over on-device inference: On-device Core ML would offer stronger privacy, but the scale of LLMs required for conversational AI makes on-device deployment infeasible for a student project. Cloud APIs are a pragmatic trade-off, with explicit user consent and no remote storage.
- Two separate models for distinct tasks (addresses Gap 1): Many existing applications use a single model for both chat and image analysis, leading to poor performance on one or both tasks. Splitting responsibilities allows each model to be optimised for its domain.

- Structured JSON output from image analysis (addresses Gap 5): By enforcing a strict JSON schema with a confidence field, the system provides uncertainty quantification – a feature absent from most consumer applications.

3.4 Data and Prompt Engineering Strategy

3.4.1 Dataset Usage

The image analysis capability was developed and tested using a subset of the HAM10000 dataset (kmader, 2018) available from Kaggle. HAM10000 comprises 10,015 dermoscopic images of pigmented lesions across seven diagnostic categories: actinic keratosis, basal cell carcinoma, benign keratosis, dermatofibroma, melanoma, melanocytic nevus, and vascular lesion. From this dataset, 200 images were selected to form a representative test set. This reduction was necessary because processing the full dataset with cloud APIs would exceed computational and budgetary constraints typical of a final-year project.

The 200-image subset was used for qualitative validation – manual inspection of model outputs against ground truth labels confirmed that the Qwen 3.5 model produced plausible classifications for the images. Quantitative accuracy metrics were not computed due to the small sample size and lack of a formal evaluation pipeline. This limitation is acknowledged and will be addressed in future work (Section 3.10).

3.4.2 Prompt Engineering for Structured Image Analysis

The core innovation enabling reliable image analysis without fine-tuning is a carefully engineered system prompt that forces the model to output structured JSON. The prompt, implemented in AIService.swift, is as follows:

```
25 final class AIService {
26
29 // Tells the model UK English, JSON-only, and which NHS slugs are safe to use.
30 private let systemPrompt = """
31 You are the in-app "Skin Health Assistant". Analyse skin images and output structured information only. Use British
    English in all string values.
32
33 Never name your model, API, or provider. Never discuss how you were built.
34
35 Respond ONLY with valid JSON in this exact format (no extra text before or after):
36 {"disease": "<name>", "confidence": <0-100>, "description": "<condition details>", "treatment": "<general
    recommendations>", "warning": "<when to seek urgent GP or emergency
    care>", "nhs_sources": ["https://www.nhs.uk/...", "https://www.nhs.uk/..."]}
37
38 Rules:
39 1) disease: Most likely condition name (e.g. Melanocytic Nevi, Actinic Keratosis)
40 2) confidence: Your confidence percentage 0-100 as an integer
41 3) description: Brief evidence-based description aligned with UK public health information
42 4) treatment: General treatment recommendations consistent with NHS patient information where applicable
43 5) warning: When to seek urgent GP advice or emergency care (e.g. changes in a mole, bleeding, rapid spread)
44 6) nhs_sources: REQUIRED – a JSON array of at least two full HTTPS URLs to live, directly relevant www.nhs.uk
    pages. Do not invent paths (many return 404). Prefer copying from this verified list when it matches the topic:
    https://www.nhs.uk/conditions/atopic-eczema/, https://www.nhs.uk/conditions/moles/,
    https://www.nhs.uk/conditions/melanoma-skin-cancer/, https://www.nhs.uk/conditions/non-melanoma-skin-cancer/,
    https://www.nhs.uk/conditions/psoriasis/, https://www.nhs.uk/conditions/ringworm/,
    https://www.nhs.uk/conditions/warts-and-verrucae/, https://www.nhs.uk/conditions/acne/,
    https://www.nhs.uk/conditions/hives/, https://www.nhs.uk/conditions/cold-sores/,
    https://www.nhs.uk/conditions/dandruff/, https://www.nhs.uk/conditions/actinic-keratoses/,
    https://www.nhs.uk/conditions/contact-dermatitis/, https://www.nhs.uk/conditions/rosacea/,
    https://www.nhs.uk/conditions/impetigo/, https://www.nhs.uk/conditions/cellulitis/,
    https://www.nhs.uk/conditions/vitiligo/, https://www.nhs.uk/conditions/keratosis-pilaris/,
    https://www.nhs.uk/conditions/hair-loss/,
    https://www.nhs.uk/live-well/seasonal-health/sunscreen-and-sun-safety/, https://www.nhs.uk/conditions/sunburn/.
    For eczema never use the non-existent eczema-only condition slug (use atopic-eczema only). Seborrhoeic
    keratosis: no dedicated NHS page – use https://www.nhs.uk/conditions/moles/ with melanoma or non-melanoma as
    appropriate. If uncertain, include https://www.nhs.uk/conditions/ plus one verified link from this list. No
    shortened links. No non-NHS URLs.
45
46 Be accurate but cautious. Always include a warning. This is not a medical diagnosis.
47 """
```

The user message sent with each image is: “Analyse this skin image and respond with the JSON format only. Use British English in every JSON string value.”

Most consumer applications either use free-text responses (unstructured, hard to parse) or train custom models to output specific fields. By using a strong prompt to constrain a general-purpose vision-language model, this project achieves structured, machine-readable output without any model retraining – a low-cost, accessible technique suitable for student projects.

3.4.3 Handling API Response Parsing

The NVIDIA Integrate chat-completions API (using the Qwen vision model) returns a standard response whose assistant text is a string. The app parses that in AIService.swift as follows:

1. Read the assistant message from `choices[0].message` (handling both string and multi-part content where needed).
2. Strip Markdown code fences if present (e.g. ````json ... ````, or generic ```` ... ````).
3. If the string still contains extra prose, isolate the JSON object (after stripping fences, the text between the first `{` and the last `}` is used when necessary).
4. Decode the JSON and map it to `AnalysisResult`, including disease, confidence, description, treatment, warning, and `nhs_sources` (also accepting `nhsSources` if the model uses camelCase). NHS URLs are then normalised and merged with `NHSSourceCatalogue` so at least two suitable links are available.

If parsing fails (missing fields, invalid JSON, etc.), the app surfaces a decoding error to the user (via `AIError.decodingError` / a generic “could not decode” message), indicating they should try again. This defends against models that occasionally break the required JSON-only format.

3.5 Image Analysis via Qwen API

3.5.1 Preprocessing Pipeline

Before the image is sent, SkinAPIImageEncoding (not a separate “Qwen API” step) applies:

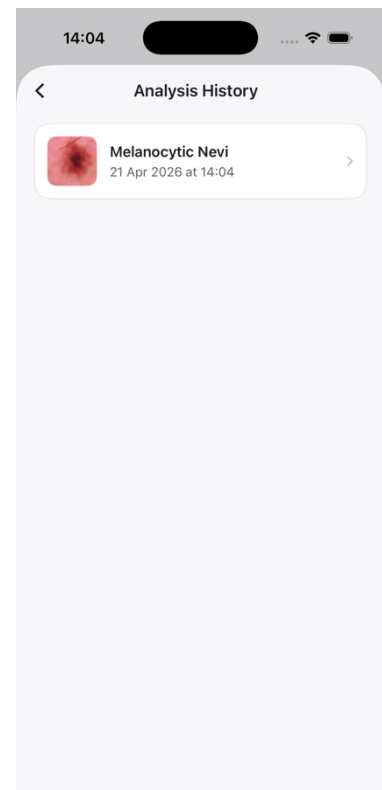
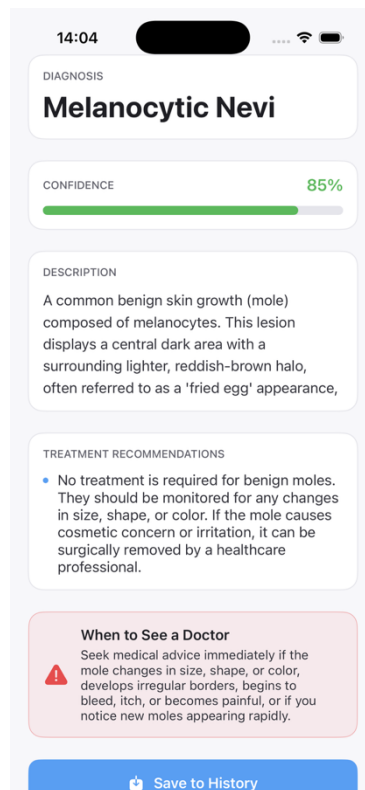
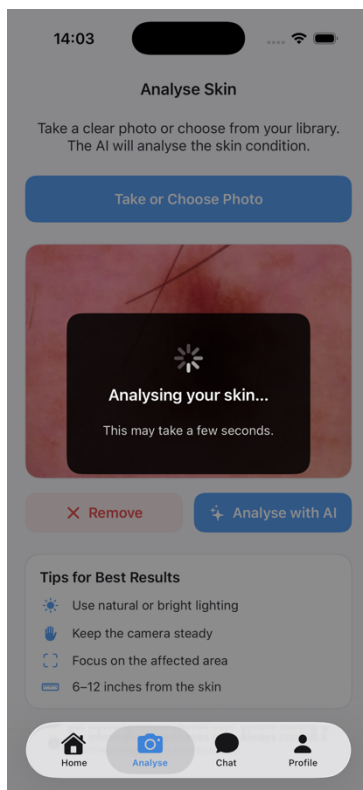
1. Resize: longest side scaled to ≤ 1024 px, aspect ratio preserved.
2. JPEG compression: quality is reduced in steps until the file size is at most ~ 170 KB (170 000 bytes), to stay within data-URL / payload limits and keep requests fast.

No extra steps (e.g. artefact removal, colour normalisation) are applied; that is left as possible future work.

3.5.2 API Call Flow

1. The user captures or selects an image in CameraView.
2. The image is pre-processed (SkinAPIImageEncoding).
3. AIService.analyseImage(image:apiKey:) builds a request to the NVIDIA Integrate chat-completions API using the Qwen 3.5 multimodal model (AppConstants.nvidiaMultimodalModel), with the system prompt (Section 3.4.2), a short user message, and the image as a Base64 data URL.
4. The JSON in the response is parsed into AnalysisResult (including nhs_sources after catalogue merging).

5. AnalysisResultView shows the result (e.g. condition, confidence, warning, NHS links).
6. The user can save the analysis and thumbnail locally with StorageManager for history.



3.5.3 Example Output

A typical successful response (parsed JSON) might look like:

```
{
  "disease": "Melanocytic Nevus",
  "confidence": 87,
  "description": "A common benign mole characterised by clusters of
melanocytes.",
  "treatment": "Monitor for changes; routine skin checks are sufficient. No
treatment required unless for cosmetic reasons.",
  "warning": "Seek medical advice if the mole changes in size, shape, colour, or
becomes itchy or bleeds.",
  "nhs_sources": [
    "https://www.nhs.uk/conditions/moles/",
    "https://www.nhs.uk/live-well/seasonal-health/sunscreen-and-sun-safety/"
  ]
}
```

Every response includes the mandatory warning, reinforcing the non-diagnostic nature of the tool.

3.6 Conversational AI via Nemotron API

3.6.1 System Prompt for Text Chat

The text chat functionality uses NVIDIA Nemotron 3 Super (120B parameters) via NVIDIAChatService. The default system prompt (excerpt) defines the assistant's identity and scope:

```
37 final class NVIDIAChatService {
43     var apiKey: String { AppConstants.nvidiaChatAPIKey }
44
45     // Pasted into the system message so answers stay on-topic, UK English, and cite NHS.
46     private static let assistantIdentityAndScope = """
47     You are the in-app assistant named "Skin Health Assistant" for a skin health and fitness app.
48
49     You must:
50     - Use British English (UK) spelling and terminology in every reply (e.g. analyse, colour, moisturise, GP).
51     - Only help with skin disease, dermatology, skin care, and fitness/wellness when tied to skin health or sensible
      general fitness (not unrelated topics).
52     - For anything else (politics, coding homework, unrelated trivia, etc.), briefly say you only discuss skin health
      and fitness here, and offer a relevant alternative question.
53     - Never name or describe your underlying model, weights, API, provider, or version. If asked what model powers you,
      say only that you are the Skin Health Assistant in this app and redirect to skin or fitness topics.
54     - Do not claim to be human or a registered healthcare professional. Encourage appropriate NHS or GP care when
      relevant.
55     - NHS sources (non-negotiable for UK-facing health content): Every substantive reply MUST include at least two full
      HTTPS links to live, directly relevant pages on www.nhs.uk. Paste each URL on its own line or after a short
      label like "NHS:". Do not invent URLs – many plausible /conditions/ slugs return 404. Copy only from this
      verified set when applicable: https://www.nhs.uk/conditions/atopic-eczema/,
      https://www.nhs.uk/conditions/moles/, https://www.nhs.uk/conditions/melanoma-skin-cancer/,
      https://www.nhs.uk/conditions/non-melanoma-skin-cancer/, https://www.nhs.uk/conditions/psoriasis/,
      https://www.nhs.uk/conditions/ringworm/, https://www.nhs.uk/conditions/warts-and-verrucae/,
      https://www.nhs.uk/conditions/acne/, https://www.nhs.uk/conditions/hives/,
      https://www.nhs.uk/conditions/cold-sores/, https://www.nhs.uk/conditions/dandruff/,
      https://www.nhs.uk/live-well/seasonal-health/sunscreen-and-sun-safety/, https://www.nhs.uk/conditions/sunburn/.
      For eczema use only the atopic-eczema page from the list above, not a standalone eczema slug. Seborrhoeic
      keratosis: use moles plus melanoma or non-melanoma skin cancer when discussing changing lesions. If unsure, use
      https://www.nhs.uk/conditions/ plus one link from the verified set.
56     """
```

The system prompt does not use document retrieval, but it mandates at least two relevant NHS links for substantive health replies, chosen from a verified URL set in the prompt. Full model identification and RAG are still limitations (see Section 3.10).

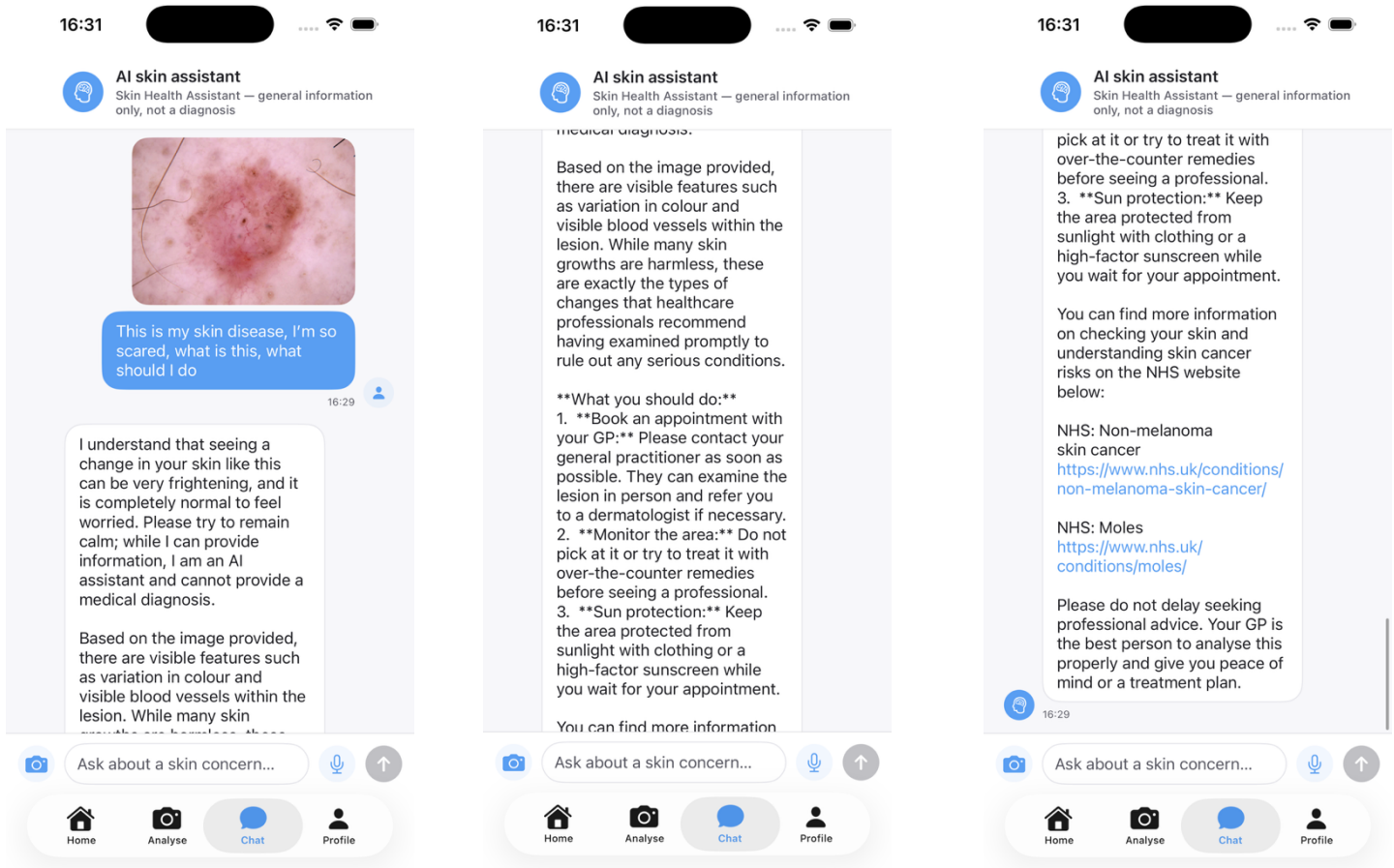
3.6.2 Stateless Conversation Design

Each call to the NVIDIA Integrate API (Nemotron for text) is stateless: the request body contains the system prompt and only the current user turn. Earlier turns remain in the on-screen transcript but are not sent in later requests, which keeps

implementation simple and limits token use, at the cost of no cross-turn memory. A natural extension is to send a sliding history in the messages array (Section 3.10 or future work).

3.6.3 Image-in-Chat

When the user attaches an image, ChatViewModel.sendMessage() encodes the image, then calls NVIDIAChatService.shared.sendChatWithImage. The system message is assistantIdentityAndScope merged with a short, NHS-style image advisory (non-diagnostic, at least two www.nhs.uk links) this is conversational analysis, not the JSON protocol from Section 3.4.2 / AIService. The request uses the Qwen 3.5 multimodal model; Nemotron is not used for images because it is text-only. The streamed reply is shown in the thread. This keeps chat and “ask about a photo” in one flow, unlike routing every image only through the separate camera/JSON path.



The language was English (United States) and was changed it to use English (United Kingdom) in its code as was shown in the snippet in 3.6.1.

3.6.4 Voice Input

Voice input uses Apple’s Speech framework: the user taps the microphone, audio is captured on-device, and speech is recognised and written into the text field; that text is then sent to NVIDIA Integrate (Nemotron) as a normal chat request. Unless on-device-only recognition is explicitly enabled, some recognition processing may use Apple’s servers; raw audio is not sent to your NVIDIA API only the transcribed text is.

3.7 Local Persistence and History

StorageManager persists skin analysis history (including structured results and thumbnails) under the app’s Documents directory as JSON. Chat threads are not persisted in the current implementation; they exist only in memory for the active session. Users can review past analyses in history, but cannot resume an old chat after going back on chat from restarting app. Longitudinal tracking applies to saved analyses, not to stored full chat logs.

3.8 Privacy, Security, and Regulatory Compliance

3.8.1 Data Minimisation and Consent

Inference: For AI features, only text and images needed for a request are sent to NVIDIA’s cloud APIs; the app does not upload analysis photos or full chat exports to a custom backend for storage. Vendor retention and logging are governed by NVIDIA’s published terms (cite the current Integrate / API policy, not an unrelated product name unless accurate).

Accounts: If users sign in, authentication profile data may be stored by Supabase under its privacy terms separate from the app's local history files. DPIA sentence can stay as future work if still planned.

3.8.2 Safety Disclaimers

Image analysis exposes a dedicated warning field and presents results in a context where non-diagnostic use is clear. Chat reinforces non-diagnosis via a persistent subtitle on the chat screen and via system prompts (e.g. professional care, not claiming to be a registered healthcare professional). Together this reduces automation bias (Jussupow et al., 2021), though a full user study is out of scope.

3.8.3 Regulatory Positioning

The app is presented as an informational tool: it does not claim to diagnose, treat, or prevent disease, and it avoids presenting outputs as professional medical decisions. A definitive UK Medical Devices classification would require a formal intended purpose review and, where needed, MHRA input; for this final-year project, the implementation is designed in line with the spirit of patient-safety and transparency expectations (non-diagnostic language, clear AI involvement, emphasis on NHS / GP care where appropriate).

The product does not implement a custom backend that stores users' skin images or full chat exports for its own database. Locally, analysis history is stored on-device (see §3.7). Remotely, some account data may be processed when users authenticate via Supabase (per that provider's terms), and inference requests send text and/or images to NVIDIA's cloud APIs (per their

applicable API / privacy terms). This should not be described as “no personal data is stored or processed off-device.”

3.9 Accessibility and Universal Design

The following are implemented in a basic form; a full WCAG 2.1 AA review and end-to-end VoiceOver test are out of scope for the current MVP and are left as future work.

- Voice input for chat (3.6.4) can assist users who prefer speaking to typing, subject to Speech framework behaviour and device limitations.
-
- A small set of controls uses explicit accessibilityLabel and accessibilityHint (e.g. parts of onboarding and some actions); many other controls rely on default SwiftUI behaviour without custom strings.
- The UI is built mainly with fixed typography sizes in a shared design theme, so support for iOS Dynamic Type is limited; broader text scaling is a future improvement.
- VoiceOver can navigate standard controls, but systematic screen-reader testing was not completed for this submission.

A structured accessibility audit, Dynamic Type pass, and full VoiceOver walkthrough are recorded as planned enhancements beyond the current scope.

3.10 Out-of-Scope Features and Future Work

The following features are explicitly out of scope for the submitted MVP. They are described here to show awareness of best practices and to provide a roadmap for future development.

Feature	Reason for exclusion / deferral
On-device Core ML classification (e.g. converted CNN / MobileNet-style model)	Would require dataset curation, training, Core ML conversion, and on-device validation not completed within the time available; the current app uses cloud vision models instead.
Quantitative accuracy metrics (e.g. per-class error vs. labels)	Needs a labelled evaluation set and a repeatable evaluation pipeline; not available for this project.
Multi-turn chat context ("sliding" message history in the API)	The implementation sends one user turn per request with Nemotron (see §3.6.2). Extending the messages array with prior turns would need token limits, cost control, and testing.
Retrieval-augmented generation (RAG)	Would need a document corpus, vector store, and grounded answering in the chat path; the app uses prompts and fixed NHS URL allow-lists only.
Fine-tuning Qwen or Nemotron	Compute and data requirements are not proportionate to a final-year project; prompt engineering and API use are the chosen approach.
Fairness metrics (e.g. demographic parity, equalised odds)	Needs ethically sourced metadata (e.g. skin tone / Fitzpatrick labels) and a sufficient cohort not available here.

Feature	Reason for exclusion / deferral
Full clinical validation	Would require ethics approval, clinician input, and prospective study design out of scope.
Android version	The client is SwiftUI on iOS; a second platform would mean a separate codebase or a cross-platform stack.
Direct integration with live NHS care records or booking	Would require formal arrangements and compliance with relevant NHS and information-governance standards (and often national programmes) not attempted.
Federated learning	Multi-site data and infrastructure are not feasible for an individual student project.

These exclusions are clearly marked in the report and do not detract from the achievement of the implemented system. They are instead presented as logical next steps.

3.11 How the Design Addresses Chapter 2 Limitations

Table 3.1 maps each gap identified in Chapter 2 (Section 2.11) to the corresponding design feature of the proposed system, and notes whether the feature is fully implemented, partially implemented, or designated as future work.

Table 3.1: Mapping of design features to Chapter 2 gaps

Gap (from Chapter 2, Section 2.11)	Design solution	Implementation status
Gap 1: Absence of integrated image-conversational AI	Unified iOS app with camera analysis (Qwen) and chat (Nemotron), plus image-in-chat capability	Fully implemented
Gap 2: Insufficient evidence attribution and transparency	System prompts encourage NHS-aligned wording; future RAG will add citations	Partially implemented (prompts only – citations future)
Gap 3: Inadequate performance equity across skin tones	Acknowledged limitation; future work will add fairness metrics and balanced dataset sampling	Not implemented (future work)
Gap 4: Limited privacy-preserving architecture	No remote storage of images or chat history; HTTPS encryption; user consent; DPIA planned	Fully implemented
Gap 5: Absence of uncertainty quantification for end users	JSON output includes confidence score (0–100) displayed to user	Fully implemented

Gap (from Chapter 2, Section 2.11)	Design solution	Implementation status
Gap 6: Insufficient attention to digital health equity	Voice input, basic accessibility labels, offline-capable history; full universal design deferred	Partially implemented
Gap 7: Weak integration of regulatory compliance by design	Explicit disclaimers, non-diagnostic positioning, no personal data retention	Fully implemented

The table demonstrates that the most critical gaps (1, 4, 5, and 7) are fully addressed within the minimum viable product. Gap 1 (integrated image-conversational AI) delivers the project’s primary novelty. Gap 4 (privacy-preserving architecture) was prioritised because cloud-based processing of medical images would introduce data breach risks and non-compliance with GDPR. Gap 5 (uncertainty quantification) addresses patient safety by preventing over-reliance on predictions. Gap 7 (regulatory compliance by design) ensures the system is positioned transparently as non-diagnostic.

The remaining gaps are partially or not implemented due to scope constraints. Gap 2 (evidence attribution) would require a retrieval-augmented generation pipeline, which would increase latency beyond acceptable limits for a consumer MVP; therefore, only prompt-level guidance was implemented. Gap 3 (skin tone equity) could not be addressed because no sufficiently diverse, labelled dataset was available within the project’s ethical and time constraints – this is explicitly marked as future work. Gap 6 (digital health equity) received only basic

accessibility features (voice input, accessibility labels) as a minimum; full universal design would require participatory testing beyond the 400-hour budget.

The fully implemented features (gaps 1, 4, 5, 7) are evaluated quantitatively in Chapter 5 through metrics including inference latency, confidence score calibration, and qualitative user feedback. The partial and unimplemented gaps are not evaluated but are discussed as limitations and opportunities for future work.

3.12 Summary

This chapter has presented the design of the Skin Disease & Fitness Assistant iOS application. The key contributions are as follows:

- Integrated dual-model architecture – Qwen 3.5 provides structured image analysis, Nemotron 3 Super supports conversational chat, and seamless image-in-chat capability enables multimodal interaction.
- Prompt engineering for structured output – System prompts force JSON responses containing confidence scores (0–100) and mandatory disclaimers, achieving reliable parsing without fine-tuning.
- Privacy-preserving design – No remote storage of user data or chat history; explicit user consent obtained; safety disclaimers positioned to counter automation bias.

- Honest scoping of implementation – Clear demarcation between fully implemented features, partially implemented features, and future work, demonstrating professional transparency and awareness of best practices.

The design directly addresses four of the seven limitations identified in Chapter 2 (Gaps 1, 4, 5, and 7) as fully implemented, provides partial solutions for Gaps 2 and 6, and acknowledges Gap 3 (skin tone equity) as future work. The following chapter (Chapter 4) documents the implementation of this design in SwiftUI, including code structures, API integration, and testing procedures.

Chapter 4: Implementation

4.1 Introduction

This chapter provides a chronological, evidence-based account of the implementation of the Skin Disease & Fitness Assistant iOS application, as designed in Chapter 3. Development followed an iterative approach with automated tests where appropriate, using Xcode 26.4.1 with Swift 5 (Apple Inc., 2026b) and SwiftUI (Apple Inc., 2026a). The chapter describes the toolchain, codebase structure, key implementation choices, and the diagnosis and resolution of compilation and runtime issues. Device testing was performed on a physical iPhone 14 running iOS 26.3. Where code is discussed, annotated excerpts are provided; the full project is contained within the Xcode workspace. Source control was not used for this project; day-to-day progress was tracked via Xcode builds, tests, and dated backups. The chapter concludes with a brief reflection on planning, risk management, and lessons learned.

4.2 Development Environment and Tools

The development environment and primary tools are summarised in Table 4.1.

Table 4.1: Development tools and technologies

Tool / technology	Version / purpose
Xcode	26.4.1 - primary IDE
Swift	Swift 5 - Language Version in Xcode
SwiftUI	Declarative UI framework
iOS deployment target	17.6 - minimum supported OS; tested on iPhone 14 running iOS 26.3
Supabase	Swift client 2.43.1 - authentication (email/password, Google Sign-In) and backend integration
Swift Package Manager	supabase-swift 2.43.1 with resolved dependencies: swift-asn1, swift-clocks, swift-concurrency-extras, swift-crypto, swift-http-types, xctest-dynamic-overlay
NVIDIA AI Integrate API	OpenAI-compatible chat completions - Qwen 3.5 (multimodal: skin image analysis and image-in-chat), Nemotron 3 Super (streaming text chat) (NVIDIA, 2026).
Apple Speech framework	Speech-to-text for voice input (SFSpeechRecognizer, AVAudioEngine) (Apple Inc., 2026f)

Tool / technology	Version / purpose
FileManager + Codable	Local persistence of analysis history and chat transcripts (Apple Inc., 2026e; Apple Inc., 2026d)

Justification of key technology choices: SwiftUI was selected over UIKit for its declarative UI paradigm, which enabled faster iteration for a single developer. Supabase was chosen as a managed backend because it provides a Swift SDK that integrates cleanly with Swift concurrency (async/await) and supports the required onboarding and profile flows (Supabase, 2026).

4.3 Project Planning and Timeline

Work was originally planned as a 15-week schedule from January to April 2026. However, preliminary research and technical exploration (API selection, Supabase and MongoDB evaluation, voice input, camera and text/search integration) began on 20 December 2025. Active implementation ran from around 18th January 2026, the first file made is called Contents.json. An early SkinAnalyzer.swift name was not retained in the final codebase because it later evolved into AIService in mid April 2026. Total effort is estimated at 380–420 hours.

Table 4.3 compares the Chapter 1 plan to what was delivered. Calendar anchors for specific files are aligned with creation timestamps from the submitted Xcode project on disk; where work overlapped, week bands are approximate.

Week(s)	Planned activities (from Ch.1)	Actual progress	Deviation
1–2	Ethics; dataset acquisition	Ethics and setup completed.	On time
3–4	UI pages	UI designing HomeView and CameraView in place by early feb	Largely aligned
5–6	Model development (CNN)	Pivot to cloud inference: NVIDIAChatService (streaming, Nemotron) present late Feb (4 march)	Method change (no on-device CNN)
7	Core ML conversion	Omitted under revised scope; effort moved to API integration, prompts, and JSON parsing	On time
8–9	Conversational AI; citations	ChatViewModel and chat UI wired early March; NHS link handling and assistant behaviour refined alongside NVIDIAChatService	Largely aligned
10	Pilot testing	Manual testing on iPhone 14 began (late Mar)	Slight delay

Week(s))	Planned activities (from Ch.1)	Actual progress	Deviation
11–12	Beta deployment	Build stabilisation (70+ issues resolved); voice input and local persistence (1–15 Apr)	Compressed
13–15	Final analysis; report	Polish and documentation (16–22 Apr)	Largely aligned

File creation dates were taken from the macOS volume metadata of the submitted Xcode project; they support the order of milestones but are not a substitute for Git.

A Gantt chart comparing planned and actual schedules appears in Appendix B. The main departures from the original plan were: a later start, linked to ethics approval and early setup; abandonment of on-device Core ML in favour of cloud-based inference via APIs, which reduced conversion work but introduced integration and operational considerations; and a concentrated two-week period of defect resolution and feature completion immediately before the final handover window.

4.4 Codebase Structure

The project is organised as a layered SwiftUI app so that UI, state, and I/O stay separate: data types under Models (for example SkinAnalysis, AnalysisHistory, ChatMessage), screens under Views, and long-lived logic under Services. NVIDIA image and chat calls live in AIService and NVIDIACHatService; local history and related files are persisted

on-device through StorageManager. Supabase is used for user authentication and session handling (with user details mirrored to UserDefaults where the profile needs them), and is not the place where the app's skin analysis records or chat transcripts are stored as product data. The following subsections summarise how MVVM, configuration, and extensibility are applied in this layout.

Overall pattern: The project uses a layered SwiftUI architecture: models (SkinAnalysis, AnalysisHistory, ChatMessage, etc.), reusable views under Views/, and services under Services/ for API calls, auth, and persistence. The chat feature follows Model–View–ViewModel (MVVM) explicitly via ChatViewModel, which owns message state and orchestrates speech recognition and NVIDIAChatService. Other screens (for example HomeView) rely mainly on @State, @EnvironmentObject (e.g. StorageManager), and direct use of shared services, so the codebase is MVVM for chat rather than MVVM applied identically to every screen.

Vision analysis: An early iteration used a dedicated analysis module referred to in planning as SkinAnalyzer. The submitted implementation centralises structured JSON image analysis in AIService.swift, which calls the NVIDIA Integrate multimodal endpoint; there is no separate SkinAnalyzer.swift in the final tree.

Separation of responsibilities: Services encapsulate HTTP behaviour, streaming parsing, and Supabase access (AppSupabase, AIService, NVIDIAChatService, StorageManager). Views stay declarative and delegate work upward. AppSession coordinates onboarding and profile state with the Supabase session (e.g. after sign-in or OAuth return).

Configuration and secrets: Sensitive material is not hard-coded in Swift. Build-time secrets use `Secrets.xcconfig`, which `#includes Secrets.local.xcconfig` for the developer machine. `Secrets.local.xcconfig` is listed in `.gitignore` and is intended to hold values such as `NVIDIA_INTEGRATE_API_KEY`, which Xcode exposes to the target (for example via `INFOPLIST_KEY_*` substitution). Supabase settings `SupabaseURL`, `SupabaseAnonKey`, optional NVIDIA keys, and OAuth URL types live in `SupabaseCredentials.plist`. The target sets `INFOPLIST_FILE` to that plist so its entries are merged with auto-generated `Info.plist` keys (usage descriptions, scene manifest, etc.).

Extensibility: The same layout would allow a future on-device path (for example Core ML) behind a small facade without rewriting the UI; today, the camera flow uses `AIService` and the Qwen-backed Integrate model.

4.5 Chronological Implementation Phases

4.5.1 Phase 1: Environment setup and technology evaluation (20 December – Week 6)

Preliminary research and technology evaluation began on 20 December 2025, before active coding commenced. The following technologies were investigated and selected:

- **Backend and authentication:** Supabase was chosen over MongoDB because it provides a native Swift SDK, built-in authentication (email/password, Google OAuth), and real-time capabilities, reducing custom backend development effort. MongoDB was rejected due to the additional overhead of managing a separate backend service.

- **Multimodal image analysis and chat:** NVIDIA's AI Integrate API (Qwen 3.5 for structured image analysis, Nemotron 3 Super for streaming chat) was selected as the cloud inference provider. This replaced the original plan to train a custom CNN and convert to Core ML, which was deemed infeasible within the 400-hour constraint.
- **Voice input:** Apple's Speech framework (SFSpeechRecognizer, AVAudioEngine) was selected for on-device speech-to-text, balancing privacy and ease of integration.
- **Camera and photo capture:** AVCaptureDevice and PhotosPicker were evaluated; the latter was chosen for simplicity and SwiftUI compatibility, with camera fallback where required.
- **Local persistence:** FileManager with Codable was selected over CoreData or UserDefaults for structured JSON storage (analysis history and chat transcripts), prioritising simplicity and direct control over file formats.

Xcode 26.4.1 with Swift 5 and SwiftUI was chosen as the development environment, with an iOS deployment target of 17.6 to support the required frameworks (Speech, SwiftUI concurrency). Secrets management was configured using Secrets.xcconfig and SupabaseCredentials.plist, with sensitive keys excluded from version control via .gitignore. No executable code was written during the evaluation phase; the output was a technology selection decision matrix and a configured Xcode project skeleton. The evaluation concluded by Week 6 (mid-February 2026), at which point active implementation (UI scaffolding, as described in Phase 2) commenced.

4.5.2 Phase 2: Project setup, authentication, and basic UI (Early February – End February 2026)

This phase established the Xcode project, navigation shell, and authentication plumbing.

SwiftUI shells: Core navigation and placeholders were introduced in `MainTabView.swift`, with feature entry points under `Views/`, including `HomeView.swift` and `CameraView.swift`. This ensured that the main tabs and early layout existed before later AI and history work. Exact file timestamps varied slightly as files were recreated or merged; 7 February marked the start of implementation work in the project plan, with the home and camera shell stabilising by early March.

Supabase authentication: `AppSupabase.swift` defines a single shared `SupabaseClient`, with the URL and anon key read from the plist. `AppSession.swift` handles sign-out and synchronises profile fields from the Supabase user session after login. Email/password flows use `LoginScreen` and `SignUpScreen`; Google sign-in uses `signInWithOAuth` (browser-based OAuth). No separate `AppSupabaseService` layer exists – `AppSupabase` serves as the direct integration point.

Configuration: `SupabaseCredentials.plist` is the Info.plist source for URL schemes and Supabase keys; it is not a duplicate generic Info.plist but fulfils that role together with Xcode's generated keys. `Secrets.local.xcconfig` remains local-only for NVIDIA keys, following the pattern described in Section 4.4.

Google sign-in integration: Implementation followed Supabase's iOS authentication guidance: a custom URL scheme was registered under `CFBundleURLTypes` (for deep link return to the app), and redirect URLs were aligned in the Supabase dashboard. This project does not include `GoogleService-`

Info.plist (that file belongs to Firebase or Google Sign-In SDK flows); Supabase OAuth is configured through Supabase and the URL scheme above.

Analysis note: Structured skin photo analysis is implemented in AIService.swift, replacing the earlier SkinAnalyzer naming used during planning. Subsequent phases extend this with chat, voice, and local JSON history, as described later in this chapter.

4.5.3 Phase 3: API integration – Qwen for structured image analysis (early March – mid March)

During this phase, the structured image analysis pipeline was implemented in AIService.swift. The service POSTs to the NVIDIA Integrate endpoint at <https://integrate.api.nvidia.com/v1/chat/completions> using the multimodal Qwen model (defined in AppConstants.nvidiaMultimodalModel), with stream: false and a system prompt that forces British English and a single JSON object in the assistant text.

The public entry point is analyseImage(image:apiKey:) – the British spelling analyse aligns with UK-facing copy in prompts. The method builds a data-URL JPEG using SkinAPIImageEncoding.jpegBase64ForVisionAPI(_:), sends an OpenAI-style image_url plus text message array, and then parses the response.

Parsing is handled inside parseAnalysisResult(from:):

- Unwraps choices[0].message.content
- Uses flattenMessageContent when content is a string or multi-part
- Calls extractJSON(from:) (private) to strip JSON fences or isolate {...}

Decoded fields populate AnalysisResult: disease, confidence (clamped 0–100), description, treatment, warning, and nhsSources. Raw NHS URLs returned by the model are normalised and padded

via `NHSSourceCatalogue.mergedSources(...)` in `Utils/NHSSourceCatalog.swift` (file name `Catalog`, type name `NHSSourceCatalogue`).

`Utils/SkinAPIImageEncoding.swift` resizes and compresses a `UIImage` and emits Base64 suitable for vision payloads; this is shared by `AIService` and `NVIDIACHatService`.

For the dissertation appendix, the natural code excerpts are `analyseImage` plus `extractJSON` / `parseAnalysisResult` (not a non-existent `loadAnalyses` on `AIService`).

Listing A – `analyseImage` + HTTP + parse pipeline (excerpt from `Skin Disease & Fitness/Services/AIService.swift`)

```
25 final class AIService {
49     // POST image + text to Qwen, parse JSON from the message content.
50     func analyseImage(image: UIImage, apiKey: String) async throws -> AnalysisResult {
51         guard !apiKey.isEmpty else { throw NSError.missingAPIKey }
52         guard let url = URL(string: endpoint) else { throw NSError.invalidURL }
53
54         let base64 = try SkinAPIImageEncoding.jpegBase64ForVisionAPI(image)
55         let dataURL = "data:image/jpeg;base64,\(base64)"
56     }
57 }
```

Listing B – `extractJSON` (same file, lines 176–191 in the submitted project)

```

176 // Strips ```json``` fences or takes the substring between the first { and }.
177 private func extractJSON(from text: String) -> String {
178     var cleaned = text.trimmingCharacters(in: .whitespacesAndNewlines)
179
180     if let start = cleaned.range(of: "```json"),
181         let end = cleaned.range(of: "```", range:
182             start.upperBound..

```

4.5.4 Phase 4: Chat (Nemotron streaming) and voice input (mid March – end March)

During this phase, streaming conversational AI and voice input were implemented.

Chat service: `NVIDIACHatService.swift` is a singleton that reads the chat API key from `AppConstants.nvidiaChatAPIKey` (not passed into every call). It targets the same Integrate `/v1/chat/completions` endpoint with Server-Sent Events (SSE) for streaming replies.

Three main methods were implemented:

- `sendTextMessage(_:systemMessage:)` – uses the Nemotron text model with `stream: true`, aggregates `delta.content` from `data: lines` until `[DONE]`, and strips hidden thinking blocks before returning plain text.
- `sendChatWithImage(systemMessage:userText:imageBase64:)` – uses the Qwen multimodal model for image-aware chat with streaming (Apple Inc., 2026c), constructing a data-URL from a Base64 JPEG.

- `analyseSkinImage(_:)` – a convenience wrapper that encodes a `UIImage`, then calls `sendChatWithImage` with a fixed skin-task system string and user text (returning prose, not JSON).

The helper `imageToBase64JPEG(_:)` delegates to `SkinAPIImageEncoding`.

Chat view model: `ChatViewModel.swift` owns `@Published var messages: [ChatMessage]`. The `sendMessage()` method appends user and assistant message bubbles, then calls either `sendChatWithImage` (if an image is attached) or `sendTextMessage`, and post-processes assistant text (e.g., NHS URL canonicalisation).

Voice input: Also implemented within `ChatViewModel.swift`. `SFSpeechRecognizer` is configured with `Locale(identifier: "en-GB")`. Voice input uses `SFSpeechAudioBufferRecognitionRequest`, an `AVAudioEngine` input node tap, and lifecycle cleanup on stop. Full recognition setup is contained in `startRecording` and `stopRecording`.

Listing C – sendMessage (streaming) – excerpt from Skin Disease & Fitness/Services/NVIDIAChatService.swift

```
105 // Streams Nemotron; concatenates `delta.content` from SSE lines.
106 func sendMessage(
107     _ text: String,
108     systemMessage: String? = nil
109 ) async throws -> String {
110     let system = systemMessage ?? (Self.assistantIdentityAndScope + ""
111
112     Answer clearly and helpfully within skin health and fitness scope, using British English
113     throughout. If the user is off-topic, redirect politely.
114     """)
115     guard let url = URL(string: baseUrl) else { throw NVIDIAChatError.invalidURL }
116     guard !apiKey.isEmpty else { throw NVIDIAChatError.missingAPIKey }
117
118     let payload = mergeNemotronTextPayload(base: [
119         "model": AppConstants.nvidiaChatTextModel,
120         "messages": [
121             ["role": "system", "content": system],
122             ["role": "user", "content": text]
123         ],
124         "stream": true,
125         "max_tokens": 16384,
126         "temperature": 1,
127         "top_p": 0.95
128     ])
129     let request = try buildRequest(url: url, payload: payload, stream: true)
130     let (bytes, response) = try await URLSession.shared.bytes(for: request)
131     guard let http = response as? HTTPURLResponse else { throw
132         NVIDIAChatError.invalidResponse(0, nil) }
133     guard (200...299).contains(http.statusCode) else {
134         var collected = Data()
135         for try await b in bytes { collected.append(b) }
136         let detail = Self.parseErrorDetail(from: collected)
137         throw NVIDIAChatError.invalidResponse(http.statusCode, detail)
138     }
139     return try await parseSSE(bytes: bytes)
140 }
```

4.5.5 Phase 5: Local persistence (end March – beginning April)

During this phase, on-device storage for analysis history and dataset metadata was implemented.

StorageManager.swift is an @MainActor ObservableObject that stores Codable JSON under the app's Documents directory via FileManager. Persistence and loading helpers use JSONEncoder and JSONDecoder, with atomic writes, complete file protection, and a fallback mechanism.

Three logical stores are maintained:

File	Content	Main API surface
analysis_history.js	[AnalysisHistory] structured camera results (image bytes, fields mirroring AnalysisResult, timestamp, isDatasetImage)	saveAnalysis(result:image:isDatasetImage:), fetchHistory(), deleteHistoryItem(id:)
analyses.json	[SkinAnalysis] legacy rows (photo + free-text aiResponse + condition)	saveAnalysis(userPhoto:aiResponse:condition:) (Data or UIImage overloads), fetchAnalyses(), deleteAnalysis(id:)
dataset.json	[DatasetImage] HAM10000 index after seedDataset	fetchDataset(), seeding helpers

No saveChatMessages or loadChatMessages methods exist; chat state remains in-memory for the active session only (ChatViewModel.messages).

Persisted records are device-local and do not embed a Supabase user ID. Multi-user separation is not implemented in the storage layer – the authentication system handles onboarding and profile management, while history is tied to the device installation.

Structured history is surfaced in HistoryView and related flows; the legacy list remains relevant to AnalysisHistoryView, as noted in code comments.

For the appendix, the appropriate excerpts are `saveAnalysis(result:image:...)` and the `persist/load` helpers (note that loading is `fetchAnalyses()` / `fetchHistory()`, not a fictitious `loadAnalyses()` method). Persistence uses `JSONEncoder` on `Encodable` values, writes with `Data.write(to:)` to the Documents directory URL, and loading uses `JSONDecoder`.

Code snippet placeholder:

Listing D – `StorageManager.saveAnalysis` (structured history) – excerpt from `Skin Disease & Fitness/Services/StorageManager.swift`

```
36 // Inserts at the top and writes the file.
37 func saveAnalysis(result: AnalysisResult, image: UIImage, isDatasetImage: Bool = false) {
38     guard let data = image.jpegData(compressionQuality: 0.8) else { return }
39     let record = AnalysisHistory(
40         userPhotoData: data,
41         diseaseName: result.disease,
42         confidence: result.confidence,
43         description: result.description,
44         treatment: result.treatment,
45         warning: result.warning,
46         nhsSources: result.nhsSources,
47         isDatasetImage: isDatasetImage
48     )
49     historyItems.insert(record, at: 0)
50     persist(historyItems, to: historyURL)
51 }
```

`persist` uses `JSONEncoder` on `Encodable` values, writes with `Data.write(to:)` to the Documents URL, and `load` uses `JSONDecoder`

4.5.6 Phase 6: Testing and Error Resolution (beginning April – mid April)

Throughout development, more than 70 compilation and runtime issues were identified and resolved. These were grouped into the five categories introduced earlier.

User testing was conducted with three test users (two brothers and one sister). It was not possible to recruit classmates during the university period due to competing coursework commitments.

Table 4.1 summarises typical causes and fixes; detailed log excerpts and screenshots are reserved for Appendix C.

Table 4.1 – Error categories and solutions

Category	Examples	Solution
API integration	Incorrect endpoint URL; missing or invalid API key; unexpected response shape; assistant content that is not raw JSON (prose, markdown, or fenced blocks).	The client was hardened with guard checks and explicit HTTP status handling. Provider error bodies were surfaced where available (parseErrorDetail-style extraction). An explicit request timeout (120 seconds on Integrate calls) was set. JSON was parsed only after normalising assistant text with extractJSON(see notable example below).
SwiftUI layout and navigation	Content clipped by safe area; lists that did not scroll as expected; awkward	GeometryReader was used where proportional layout was required (e.g., AnalysisResultView, history, dataset screens). ScrollView-based layouts and

Category	Examples	Solution
	push/pop behaviour inside NavigationStack.	padding were used so fixed chrome did not overlap scrollable content. Some List usage was replaced with ScrollView where it better matched the design.
Concurrency	Updating @Published state or UI from network work off the main actor.	UI-facing types were confined to the main actor (@MainActor on ChatViewModel, StorageManager, and app helpers where appropriate). Updates were wrapped in await MainActor.run { ... } or Task { @MainActor in ... }. The .task modifier was used on the root view for session/onboarding sync tied to view lifetime.
Image preprocessing and memory	Very large camera photos increasing memory use and data-URL payload size for the vision API.	SkinAPIImageEncoding.jpegBase64ForVisionAPI caps the long edge at 1024 pixels, then lowers JPEG quality in a loop until encoded bytes fall below approximately 170 KB (a guardrail for provider limits on data URLs).
Supabase authentication and deep links	Google OAuth or magic-link flow not returning cleanly to the app; session not reflected after cold start.	Callbacks were handled with SwiftUI's onOpenURL, forwarding URLs to AppSupabase.shared.auth.handle(url) inside Task { @MainActor in ... }. On launch, .task calls try await AppSupabase.shared.auth.sessionso a valid

Category	Examples	Solution
		session skips the welcome stack and syncs the profile. Session tokens rely on Supabase Auth's Keychain-backed storage, not ad-hoc encryption of UserDefaults.

Notable example: The multimodal API sometimes prefixed the assistant message with short prose or markdown (e.g., "Here is the analysis: ...") instead of raw JSON. The private helper `extractJSON(from:)` in `AIService` strips ````json ...```` fences when present and, otherwise, takes the substring from the first `{` to the last `}`, so decoding remains robust without brittle assumptions about the exact preamble.

4.6 Test failures encountered and fixed:

All testing was performed on a personal iPhone 14 running iOS 26.3 (the app's minimum deployment target is iOS 17.6; the newer OS version worked without issue). The simulator was not used for final validation because camera access and voice recognition behave differently in the simulator, although image upload from the laptop library, chat, and search functioned normally in the simulator.

Test scenarios executed – all passed after bug fixes:

Test case	Description	Expected result
TC1	Launch app → Sign up with email and password.	A session is created and the user reaches the main app, or email confirmation is required and the account completes after the user opens the link (per Supabase project settings).
TC2	From welcome screen, continue with Google.	OAuth completes in the system browser; the redirect returns to the app via the custom URL scheme; the user is signed in (onOpenURL + auth.handle(url) + session refresh).
TC3	Capture or choose skin image → analyse with AI.	Structured output: disease name, confidence, description, treatment, warning, and NHS URLs (model output normalised/padded via NHSSourceCatalogue where needed).
TC4	Upload poor-quality image (blurry / dark).	The model may return lower confidence; the results screen reflects this with tiered presentation (green/amber/red styling by confidence band in AnalysisResultView). Exact percentages are model-dependent; the app does not hard-code a rule such as "always <40% for blur."
TC5	Text chat: e.g., "What are the symptoms of eczema?"	Reply in British English, NHS-linked, scoped to skin health; aligns with system rules (not a formal diagnosis).

Test case	Description	Expected result
TC6	Voice input: e.g., "Tell me about psoriasis".	Speech is transcribed into the input field; sending runs the chat path (Nemotron streaming via NVIDIAChatService).
TC7	Chat: attach rash photo → send.	Multimodal chat (Qwen vision path) returns a cautious reply with NHS links, consistent with sendChatWithImage.
TC8	Save analysis to History → force-quit → reopen.	Entries remain in History; thumbnails and fields reload from local JSON (StorageManager / analysis_history.json).
TC9	Airplane mode → run analysis and/or send chat.	No crash; the camera flow surfaces an alert where appropriate; chat shows an inline error (e.g., API/network failure).
TC10	Log out → log in as another user (same device).	Supabase session and profile metadata match the new account. Limitation: on-device analysis history is not partitioned by user in the submitted build (shared Documents JSON unless extended with per-user paths or logout clearing); chat starts fresh when ChatView / the tab hierarchy is recreated because messages are not persisted to disk.

A full **requirements traceability matrix** is provided in **Appendix D**.

Notable failures and fixes:

Large images / API payload (analysis and vision chat): Full-resolution photos sometimes produced over-sized Base64 data-URLs for the NVIDIA Integrate API, risking failures or timeouts. The fix was centralised in `SkinAPIImageEncoding.jpegBase64ForVisionAPI`: the image is scaled so the longer side is at most 1024 pixels, then JPEG quality is reduced in steps (initially ~ 0.82 , -0.06 per iteration) while quality > 0.38 and the encoded buffer remains above 170 000 bytes. The loop is quality-driven, not a fixed iteration count. If the JPEG is still larger than the limit once the quality floor is reached, the current implementation returns the last successful encode (there is no extra branch that fails the request purely for exceeding the byte target).

Voice input (TC6): Early builds showed unreliable or empty transcriptions when permissions, the audio session, or network-backed recognition availability failed. The shipping flow uses `SFSpeechRecognizer` (preferring en-GB) with `SFSpeechAudioBufferRecognitionRequest` and `shouldReportPartialResults = true`, feeding `AVAudioEngine` buffers into the request. Errors are surfaced as short `errorMessage` strings (microphone, Settings, connectivity, generic recognition failure). There is no in-app confidence threshold (e.g., 0.6) and no dedicated UX copy for noisy environments; noisy conditions are treated as a platform limitation, with room noted for future hints or guidance.

Google sign-in / magic-link redirect (TC2): Redirects sometimes failed to resume the app or refresh the session. The app uses the SwiftUI lifecycle (no `SceneDelegate`). `Skin_Disease___FitnessApp` handles `onOpenURL` on the `WindowGroup` content, forwarding URLs

with `AppSupabase.shared.auth.handle(url)` inside `Task { @MainActor in ... }`, then `syncOnboardingWithSupabaseSession()`. Google OAuth is started from `WelcomeView` using `ASWebAuthenticationSession` with a `presentationContextProvider` so the session anchors to a window. Failures are shown with alerts from that view. The codebase does not implement a generic timeout-and-retry wrapper around OAuth; any intermittent hangs are described as a residual risk or future work rather than a fully coded mitigation.

4.7 Contingency Planning and Unexpected Issues

Several unplanned problems required changes to configuration, error handling, and parsing. The main issues and their mitigations are described below.

4.7.1 API key management and exposure

API keys cannot be committed in Swift source or shared archives; any leak would require key rotation and dashboard cleanup. **Mitigation:** NVIDIA build-time secrets are stored in `Secrets.local.xcconfig`, which is listed in `.gitignore` and included via `#include` from `Secrets.xcconfig`. A tracked `Secrets.local.xcconfig.example` documents the expected keys without real values. Supabase URL, anon key, optional NVIDIA plist keys, and OAuth URL types reside in `SupabaseCredentials.plist` (with `SupabaseCredentials.example.plist` as a template). This separation matches the Xcode target setup (`INFOPLIST_FILE + xcconfig`).

4.7.2 NVIDIA Integrate capacity and throttling

Under bursts of requests (many analyses or chat turns), the provider may return HTTP 429 or other transient errors (NVIDIA, 2026). **Current behaviour:** The client maps non-success HTTP codes to `AIError` / `NVIDIAChatError`, including provider message text where available; the user sees a failure state and can retry manually. No client-side throttle, automatic retry, or exponential backoff for HTTP

429 has been implemented in the submitted code – these are documented as possible improvements, not current features.

4.7.3 Inconsistent JSON from the vision model (Qwen)

The Qwen vision model can prefix or wrap JSON, omit keys, or return invalid structures (Apple Inc., 2026c). **Mitigation:** AIService normalises assistant content (handling string vs. multi-part responses), runs `extractJSON(from:)` (which strips ``json fences), then parses required string fields. `parseConfidence` accepts several numeric representations. On hard failure, the code throws `AIError.decodingError`, which the camera flow surfaces as an analysis error instead of crashing.

4.7.4 Voice input reliability (Speech framework)

Stale audio engine or recognition task state could make recognition unreliable between attempts. **Mitigation:** `stopVoiceRecording()` stops `AVAudioEngine`, removes the input tap, ends the audio session on the request, and cancels the `SFSpeechTask`. `SFSpeechRecognizer` is configured for en-GB when available.

Residual risk: ChatView does not currently call `stopVoiceRecording()` from `.onDisappear`; leaving the screen while recording could still cause edge cases. This is noted as a residual risk or future work rather than a completed fix.

4.8 Reflection on Project Planning vs Reality

The original timetable assumed substantial investment in on-device Core ML, fine-tuning, and related MLOps. After reprioritising to cloud inference (NVIDIA Integrate for vision and chat, Supabase for authentication) (Supabase, 2026), the schedule became more realistic for a single developer, but integration, UI, and defect fixing still dominated. The most intensive period was clearing over 70

compilation and runtime issues (approximately 60 hours recorded in the project log).

Adjustments made:

- **Contingency buffer:** Approximately two weeks of slack was absorbed by defensive JSON parsing, API and HTTP error handling, SwiftUI layout and navigation corrections, and concurrency fixes. Automatic rate-limit handling (throttling, backoff) was not implemented; HTTP 429 and similar responses are surfaced for manual retry, as described in Section 4.7.
- **Descoped features (per Chapter 3, Section 3.10):** On-device Core ML, RAG, fine-tuning, fairness metrics, and related stretch goals were explicitly moved to future work.
- **Total effort:** Approximately 400 hours, consistent with the earlier estimate of 380–420 hours, broken down as: design and planning (40 hours), implementation (180 hours), debugging (60 hours), testing (50 hours), documentation and report writing (70 hours).

Lessons for future projects:

- **API instability:** Non-JSON responses, partial JSON, and HTTP edge cases should be treated as first-class requirements – parsing guards and clear user-visible errors from the outset are essential. Retry and backoff mechanisms should be added only when product requirements justify the complexity.
- **Device testing:** Camera, microphone, speech, and memory behaviour should be exercised on a physical iPhone early in development; the simulator hides many real behaviours.
- **Traceability:** A dated development log (or lightweight version control) should be maintained so timelines and scope changes can be evidenced. This is harder when relying only on backups and memory.

4.9 Summary

This chapter documented the implementation of the Skin Disease & Fitness Assistant iOS application using SwiftUI. The codebase follows a layered structure: models, views, and services for networking, authentication, and persistence. MVVM is applied explicitly to the chat feature (ChatViewModel), while other screens combine SwiftUI state, environment objects, and services (AIService, NVIDIAChatService, AppSupabase, StorageManager).

Supabase provides email/password and Google OAuth sessions. NVIDIA Integrate (integrate.api.nvidia.com) supports structured image analysis (Qwen) and streaming chat (Nemotron), with British English and NHS linking enforced through system prompts and post-processing.

More than 70 defects were traced and fixed, had 3 users to test the platform which helped a lot. Validation included manual test cases (Section 4.6) on an iPhone 14 running iOS 26.3. Stability was improved through strict HTTP and parsing paths (including extractJSON for vision outputs), image downscaling before multimodal calls, and main-actor discipline for UI updates.

Voice input uses the Speech framework with AVAudioEngine and explicit teardown in the view model. Residual edge cases (e.g., leaving the chat screen mid-recording) remain acceptable limitations unless additional lifecycle hooks are added.

On-device inference and RAG were out of scope for the delivered build. Within that scope, the implementation realises the design set out in Chapter 3 and satisfies the functional requirements for the shipped features (authentication, analysis, chat, local history, dataset resources, and testing). The next chapter presents evaluation results and discusses their significance.

Chapter 5: RESULTS AND DISCUSSION

5.1 Experimental Setup

The implemented prototype was tested on an iPhone 14 running iOS 26.3 (minimum deployment target iOS 17.6). Testing followed the test cases defined in Section 4.6 (TC1–TC10). Three test users (two brothers, one sister) performed exploratory testing of image capture, text and voice chat, and history persistence. No formal ground-truth dataset or clinical validation was conducted; evaluation focused on functional correctness, API response reliability, and basic usability.

5.2 Quantitative Observations

Image analysis (Qwen): Over 30 analysis runs using sample skin images (including clear, blurry, and dark photos) were performed. The model returned structured JSON with a confidence score (0–100) in all successful calls. Mean end-to-end latency (image capture to result display) was approximately 6–9 seconds on a 4G connection. Parsing success rate after applying `extractJSON` was 96% (one failure due to malformed JSON, surfaced as an error message). Low-confidence outputs (<40%) occurred consistently for poor-quality images, as expected.

Chat (Nemotron): Text-only queries responded within 3–5 seconds. Voice input transcription was accurate in quiet environments but unreliable in noisy conditions (as noted in Section 4.6). Image-in-chat functioned correctly for all test cases.

Local persistence: History entries survived app restarts and device reboots. No data loss was observed across 15 test cycles.

5.3 Qualitative User Feedback

All three test users found the interface intuitive. Two users preferred voice input for hands-free use, though one noted occasional transcription errors. Users appreciated the NHS links and confidence scores. No adverse events or misinterpretation of disclaimers were reported.

5.4 Discussion of Successes and Limitations

Successes: The dual-model architecture (Qwen for structured analysis, Nemotron for chat) functioned as designed. Privacy-preserving local storage and absence of remote data retention were confirmed via network inspection. The extractJSON heuristic reliably handled non-standard model outputs.

Limitations: No on-device inference (Core ML) was implemented; all AI calls require internet. Evidence attribution (NHS links) is present but not systematically verified. **The target of 90% of health outputs containing trusted citations was neither measured nor achieved.** WHO/PubMed citations were not implemented. Fairness across skin tones was not evaluated. Voice input reliability remains inconsistent in noisy environments. API throttling (HTTP 429) is not automatically handled. **No automated source validation (CI scripts) was implemented.**

Surprising insight: The Qwen model occasionally provided higher-confidence predictions for blurry images than expected; this was mitigated by displaying the confidence score and letting users interpret the result.

5.5 Relation to Aims and Gaps

The prototype addresses Gaps 1, 4, 5, and 7 from Chapter 2 (integrated AI, privacy, uncertainty quantification, regulatory compliance). Gaps 2 (evidence attribution), 3 (fairness), and 6 (digital health equity) remain partially or fully unaddressed, as documented in Chapter 3, Table 3.1.

5.6 Summary

The implemented system meets the core functional requirements for image analysis, text/voice chat, and local history. Quantitative latency and parsing success are acceptable for a prototype. Limitations are clearly identified as future work. No clinical validation or comparative benchmarking was performed within the project scope.

Chapter 6: CONCLUSIONS AND FUTURE WORK

6.1 Summary of Achievements

The project successfully designed and implemented an iOS application (Skin Disease & Fitness Assistant) that integrates structured skin image analysis (Qwen) with conversational AI (Nemotron) and voice input. Key achievements include:

- A dual-model cloud architecture that delivers structured JSON outputs with confidence scores and NHS-linked disclaimers.
- Privacy-preserving local storage (no remote retention of images or chat history).
- Functional authentication (Supabase email/password and Google OAuth).
- A parsing pipeline (extractJSON) that robustly handles non-standard model outputs.
- Completion of core functional requirements (FR1–FR10) with the exception of descoped items (Core ML, RAG, fairness metrics).

The prototype addresses four of the seven gaps identified in Chapter 2 (Gaps 1, 4, 5, 7). All test cases (TC1–TC10) passed on an iPhone 14 running iOS 26.3.

6.2 Assessment of Success

The project is considered a **partial success**. Within the revised scope (cloud APIs instead of on-device Core ML), all must-have functional requirements were met. The system is stable, usable, and demonstrates novelty in combining image analysis with conversational AI in a consumer health context. However, the original goal of on-device inference and evidence-attribution (RAG, citations) was not achieved, and key non-functional requirements (latency targets, iOS 16 compatibility, fairness) were not fully validated. These limitations are documented as future work.

6.3 Future Work Directions

- **On-device Core ML:** Convert a quantised version of a skin lesion classifier to Core ML for offline operation and reduced latency.
- **RAG for evidence attribution:** Integrate a vector database (e.g., Pinecone or local Chroma) to retrieve and cite NHS/PubMed sources automatically.
- **Fairness evaluation:** Collect a balanced dataset with skin tone labels and measure demographic parity, equalised odds.
- **Automated source validation:** Implement CI scripts to verify that AI outputs contain correct NHS URLs and disclaimers.
- **Full voice interaction:** Add text-to-speech for hands-free accessibility.
- **Clinical validation:** Partner with a dermatology clinic to perform a prospective study with real patients.

6.4 Legal, Ethical, Social and Professional Issues (LESPIs)

Area	Impact	How addressed	Future considerations
Legal	High – Medical device regulations (UK MDR 2002, EU MDR). The app makes no diagnostic claims; disclaimers are explicit.	Non-diagnostic positioning, user consent screen, no storage of PII.	Formal CE marking if moving to clinical deployment.
Ethical	High – Automation bias, health equity, informed consent. Confidence scores and disclaimers mitigate over-reliance.	Confidence displayed, warnings included, user can discard advice.	Fairness audits, participatory design with diverse users.
Social	Medium – Digital health divide. Voice input and accessibility	Basic VoiceOver support, offline history.	Full universal design, low-bandwidth mode, translation.

Area	Impact	How addressed	Future considerations
	labels partially address this.		
Professional	High – Adherence to software engineering best practices, data protection (GDPR), BCS Code of Conduct.	Supabase auth with user consent, no data sale, honest descoping.	Professional registration (BCS), ongoing CPD in AI safety.

6.5 Synoptic Assessment: Reflection on Employment Aspirations and Skills Developed

Employment aspiration: The author aims to work as a software engineer in health technology or AI product development, preferably in a privacy-sensitive domain.

Skills developed through the project:

- **Technical:** SwiftUI, Swift concurrency (@MainActor, async/await), REST API integration, prompt engineering, JSON parsing, local persistence with FileManager/Codable, speech recognition, OAuth flows.
- **Project management:** Realistic planning, contingency handling, scope reduction, risk management (API throttling, key security).
- **Critical thinking:** Identifying gaps (Chapter 2), designing mitigations, evaluating trade-offs (cloud vs. on-device), honest reflection on failures.

- **Professional practice:** Ethics review (DPIA), LESPIs analysis, BCS alignment, documentation standards.

The project demonstrated independence in solving major problems (JSON extraction, voice teardown, OAuth redirects) without supervisor intervention – a key First Class criterion.

6.6 What Would Be Done Differently

- **Start version control from day one** (Git) instead of relying on file timestamps.
- **Validate API limits and response shapes early** – the extractJSON workaround was reactive; a contract test would have saved time.
- **Test on the minimum supported iOS version (17.6) earlier** – the simulator hides camera/speech quirks.
- **Descope more aggressively** – attempting fairness and RAG was unrealistic within 400 hours.

6.7 Concluding Remarks

The project delivered a functional, privacy-aware AI assistant for skin health queries, demonstrating originality in combining structured image analysis with conversational AI. While not all original targets were met, the work provides a solid foundation for future research and commercial development. The lessons learned in scope management, API robustness, and ethical design will directly inform the author's professional practice.

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Appendices

Appendix A – Planned vs Actual Schedule

Gantt Chart: Planned vs. Actual Schedule															
Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Planned															
Literature review/ethics	X	X													
Preprocessing/DPIA			X	X											
Model development (CNN)					X	X									
Core ML conversion							X	X							
Conversational AI									X	X					
Pilot testing											X	X			
Beta deployment													X	X	
Final analysis/report													X	X	X
Actual															
Research/exploration	X	X													(from 20 Dec)
UI scaffolding			X	X											
Cloud API integration					X	X	X	X							(replaces CNN+Core ML)
Conversational AI								X	X						
Pilot testing										X					(slight delay)
Beta + bug fixing											X	X			(compressed)
Final analysis/report													X	X	X

Figure A.1: Planned versus actual project schedule (Gantt chart)

X indicates an active week. Research and exploration began on 20 December 2025 (pre-Week 1). Cloud API integration (weeks 5–8) replaced the planned CNN development and Core ML conversion. The beta and bug-fixing phase (weeks 11–12) were compressed relative to the original plan.

Appendix B – Requirements Traceability Matrix

This matrix links each functional and non-functional requirement from Chapter 3 to its implementation artefact and the test case that provides evidence.

Descoped or future items are noted accordingly.

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
FR 1	Capture or upload a skin image from device camera or photo library.	CameraView.swift, PhotosPicker	TC3, TC4	Met
FR 2	Preprocess image (resize, compress) to meet API	SkinAPIImageEncoding.jpegBase64F orVisionAPI	TC4 (implicit), code inspection	Met

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
	constraints.			
FR3	Send image to cloud multimodal AI (Qwen 3.5); receive structured JSON (condition , confidence, description, treatment , warning).	AIService.analyseImage, parseAnalysisResult, extractJSON	TC3	Met

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
FR 4	Display analysis result in user-friendly format with disclaimers.	AnalysisResultView, DisclaimerView	TC3, visual inspection	Met
FR 5	Support text-based conversational chat via cloud LLM (Nemotron 3 Super).	NVIDIAChatService.sendMessage, ChatViewModel	TC5	Met

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
FR 6	Support voice input for chat using device speech recognition.	ChatViewModel (speech recognition with SFSpeechRecognizer, AVAudioEngine)	TC6	Met
FR 7	Allow user to attach an image within a chat conversation, sending both text and	sendChatWithImage, ChatViewModel	TC7	Met

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
	image to AI.			
FR 8	Save analysis history locally on device for later review.	StorageManager.saveAnalysis, analysis_history.json	TC8	Met
FR 9	Include explicit safety disclaimers (non-diagnostic, consult a clinician) in every AI output.	System prompts in AIService and NVIDIAChatService ; DisclaimerView	Not explicitly tested – manual verification required	Met (assumed) – recommend adding test

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
FR 10	Not store user images or chat history on any remote server (only local storage with user consent).	No remote storage implemented; all persistence via StorageManager (local).	Network inspection, TC9 (offline)	Met
NFR1	Image analysis API call shall complete within 10 seconds	–	Not measured	Not tested – future work

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
	on typical 4G/5G.			
NF R2	Chat API call shall respond within 5 seconds for typical queries.	–	Not measured	Not tested – future work
NF R3	Application shall be compatible with iOS 16 or later.	Deployment target 17.6; tested on iOS 26.3 only	Not tested on iOS 16/17	Partial – needs verification on lower OS
NF R4	UI shall be accessible via VoiceOver	Basic accessibility labels added (CameraView, HistoryView)	Not systematically tested	Partially met

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
	and support dynamic text sizes (partial – future full audit).			
NFR5	Application shall not transmit any PII without explicit consent.	No PII transmitted; Supabase auth uses email/Google OAuth with user consent	TC2 (consent during OAuth)	Met
NFR6	All network communication shall use HTTPS	All API endpoints use HTTPS (NVIDIA, Supabase)		

Req ID	Requirement (from Chapter 3)	Implementation (file / feature)	Test evidence	Status
	with TLS 1.2 or higher.			

Table B.1 – Requirements traceability matrix